

From Quantum Error Correction to Fault-Tolerant Quantum Classification: A Simulation Study on CIFAR-10

Abstract

This report presents a step-by-step simulation study of quantum error correction (QEC) and fault-tolerant quantum computing (FTQC) in the context of a hybrid quantum-classical classifier for CIFAR-10. The study begins with elementary repetition-code experiments and fault-propagation tests, then progresses to circuit-level noisy QEC, general Pauli noise, the $[[5, 1, 3]]$ perfect code, flagged stabilizer extraction, syndrome-history decoding, and a gate-aware logical fault model. The underlying eight-logical-qubit classifier uses learned 32-dimensional ResNet18 features with data re-uploading and achieves an ideal test accuracy of 91.43% with a macro-F1 score of 91.46%. Ideal QEC substantially suppresses logical errors and preserves task accuracy, but circuit-level noise reveals that QEC overhead can itself dominate the computation. In the $[[5, 1, 3]]$ study, replacing naive noisy stabilizer extraction with a flagged construction reduces the logical error floor at zero external data noise from 3.9094% to 1.3274%, a reduction of approximately 66%. However, three complete noisy syndrome rounds increase the mean logical error because the added extraction circuitry introduces additional faults. The final gate-aware model selects the adaptive flagged decoder as the best fixed decoder, but still predicts 4.21 expected non-identity logical fault events per inference at zero external data noise under the chosen gadget-overhead assumptions. Consequently, the final fault-tolerant proxy reaches only 23.52% CIFAR-10 accuracy at this operating point and does not meet the target accuracy of 77%. These results demonstrate that ideal-code capability alone is insufficient: practical fault tolerance depends critically on syndrome-extraction design, decoder strategy, logical-gate overhead, and the placement frequency of error-correction gadgets.

1 Introduction

Quantum error correction is essential for reliable quantum computation because physical qubits are vulnerable to gate errors, decoherence, state-preparation errors, and measurement noise. A quantum error-correcting code can suppress physical noise by encoding one logical qubit into multiple physical qubits, but error correction alone does not guarantee fault tolerance. The error-correction circuitry itself contains gates, measurements, and ancillas that can fail, and a single poorly placed fault may propagate into multiple data-qubit errors. Fault-tolerant quantum computing therefore requires not only an error-correcting code, but also circuit constructions that prevent a small number of physical faults from becoming uncorrectable logical faults.

The objective of this study is to develop that distinction experimentally rather than only conceptually. We begin with a three-qubit repetition code, where bit-flip errors, syndrome extraction, recovery, and repeated measurements can be inspected directly. We then introduce noisy gates and measurements to study break-even behavior and compare transversal and non-transversal logical operations. After establishing these foundations, we connect the QEC study to a practical machine-learning workload: a ten-class CIFAR-10 classifier with an eight-qubit variational quantum circuit (VQC). The classifier is first evaluated under unprotected quantum noise and idealized QEC, after which increasingly realistic error-correction models are introduced.

The study ultimately focuses on the $[[5, 1, 3]]$ perfect code because, unlike the bit-flip repetition code, it can correct an arbitrary single-qubit Pauli error X , Y , or Z . We first characterize the exact ideal logical channel of the code by enumerating all $4^5 = 1024$ physical Pauli patterns. We then make stabilizer extraction noisy, introduce flag ancillas to detect dangerous hook errors, compare raw repeated syndromes with time-resolved syndrome-history decoding, and finally account for the additional logical overhead associated with arbitrary rotations and entangling gates. This progression makes it possible to identify not only when QEC helps, but also why an apparently stronger fault-tolerant construction may perform worse once its physical overhead is included.

The central finding is that the transition from ideal QEC to practical FTQC is dominated by circuit overhead. Ideal error correction preserves the classifier close to its 91.43% noiseless accuracy over a substantial noise range. In contrast, naive noisy stabilizer extraction creates a large logical-error floor even when no external data noise is applied. Flagged extraction substantially improves this behavior, but repeated noisy extraction and gate-aware logical overhead again increase the accumulated logical-fault burden. The resulting experiments provide a complete numerical narrative from ideal error suppression to a regime in which the protection circuitry itself becomes the dominant source of failure.

2 Study Design and Experimental Progression

The complete study is organized as a sequence of increasingly realistic models. The first stage establishes basic QEC using the three-qubit repetition code and verifies the theoretical logical bit-flip rate. The second stage introduces noisy syndrome measurements, circuit-level error sources, and fault propagation through logical gates. The third stage builds the CIFAR-10 hybrid quantum classifier and evaluates its sensitivity to quantum noise. The fourth stage replaces the bit-flip-only model with general Pauli depolarizing noise and the $[[5, 1, 3]]$ code. The final stages introduce noisy stabilizer extraction, flagged fault-tolerant error correction (FTEC), syndrome-history decoding, and gate-aware logical overhead.

3 Hybrid Quantum–Classical CIFAR-10 Model

3.1 Classical feature extractor

The original CIFAR-10 images contain $32 \times 32 \times 3$ input values, which is too large to encode directly into an eight-qubit circuit without aggressive compression. A preliminary eight-component PCA representation achieved only 30.67% test accuracy with a classical classifier, indicating that unsupervised linear compression discarded too much class-discriminative information. We therefore used a pretrained ResNet18 backbone followed by a learned 32-dimensional bottleneck. These 32 learned features were retained as the input representation for the quantum classifier.

3.2 Eight-qubit variational quantum circuit

The quantum model uses eight logical qubits and four rounds of data re-uploading, since $32 = 4 \times 8$. In each upload, eight features are encoded through R_Y rotations, followed by trainable R_Y and R_Z rotations and a ring of CNOT gates. The circuit measures the expectation values $\langle Z_0 \rangle, \dots, \langle Z_7 \rangle$, producing eight quantum features. A classical head maps these features through a $8 \rightarrow 32 \rightarrow 10$ network to the ten CIFAR-10 classes.

Table 1: Main stages of the simulation study.

Stage	Model	Main question
QEC foundations	Three-qubit repetition code	Can a distance-3 code suppress a single physical bit-flip error from first to second order?
Noisy syndrome	Repeated syndrome extraction	When does measurement repetition improve decoding?
Fault propagation	Transversal vs. non-transversal logical CNOT	Can a single gate fault spread into multiple errors within a code block?
Quantum ML baseline	Eight-qubit hybrid CIFAR-10 classifier	How sensitive is the trained classifier to quantum noise?
Ideal general-Pauli QEC	[[5, 1, 3]] perfect code	How strongly can ideal single-qubit Pauli correction suppress depolarizing noise?
Noisy stabilizer QEC	Ancilla-mediated stabilizer extraction	How much logical noise is introduced by the correction circuit itself?
Flagged FTEC	Syndrome ancilla + flag ancilla	Can dangerous correlated hook errors be detected and mitigated?
Temporal decoding	Three noisy flagged syndrome rounds	Does time-resolved syndrome information outperform raw majority voting?
Gate-aware model	FT Logical-gadget overhead proxy	Does protection remain beneficial after accounting for non-transversal logical-gate overhead?

The trained hybrid model reaches 91.43% test accuracy and 91.46% macro-F1 in the ideal noiseless simulation. Figure 1 shows the training and validation accuracy, while the corresponding confusion matrix and per-class results are retained for later error analysis.

4 Quantum Error-Correction Foundations

4.1 Three-qubit repetition code

For a single logical qubit encoded as

$$|\psi_L\rangle = \alpha|000\rangle + \beta|111\rangle, \quad (1)$$

the three-qubit repetition code corrects one bit-flip error. If each physical qubit experiences an independent bit flip with probability p , logical failure occurs when at least two of the three qubits flip. The corresponding logical error probability is

$$p_L = 3p^2(1 - p) + p^3 = 3p^2 - 2p^3. \quad (2)$$

At $p = 0.10$, the simulation produces an unprotected error rate of 0.0996 and a QEC error rate of 0.0283, in close agreement with the theoretical value $p_L = 0.0280$. This experiment provides the first direct demonstration that a distance-3 code can suppress the leading first-order physical error contribution.

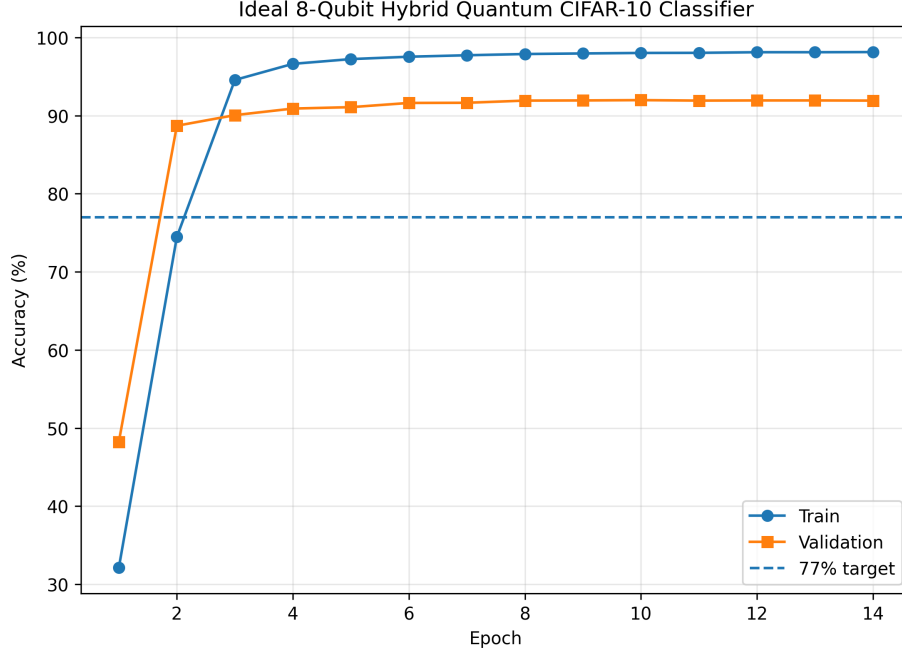


Figure 1: Training and validation accuracy of the ideal eight-qubit hybrid quantum–classical CIFAR-10 classifier. The dashed horizontal line marks the 77% target accuracy used later in the fault-tolerance study.

4.2 Repeated syndrome extraction

Syndrome bits were then made noisy. With a syndrome-bit error probability of 0.10, a single syndrome round was correct in approximately 81.14% of trials, while three repeated rounds with majority voting increased the rate to approximately 94.50%. When both data noise and syndrome noise were present, using $p_{\text{data}} = 0.05$ and $p_{\text{syn}} = 0.10$, the unprotected error rate was 0.0472, one-round QEC reduced it to 0.0232, and three-round QEC reduced it further to 0.0111. These results establish the benefit of repeated syndrome information when the underlying data error is assumed to remain fixed during repetition.

4.3 Fault propagation and transversal operations

We next compared a transversal logical CNOT with a deliberately non-transversal construction. In the transversal implementation, each physical control qubit interacts with only the corresponding target qubit, limiting the spread of a single gate fault. The measured logical error exhibits approximately second-order scaling, with a fitted log–log slope of 1.86. The non-transversal construction instead exhibits nearly linear scaling with a slope of 1.05, demonstrating how one physical fault can propagate into an uncorrectable multi-qubit error. Figure 2 visualizes this separation.

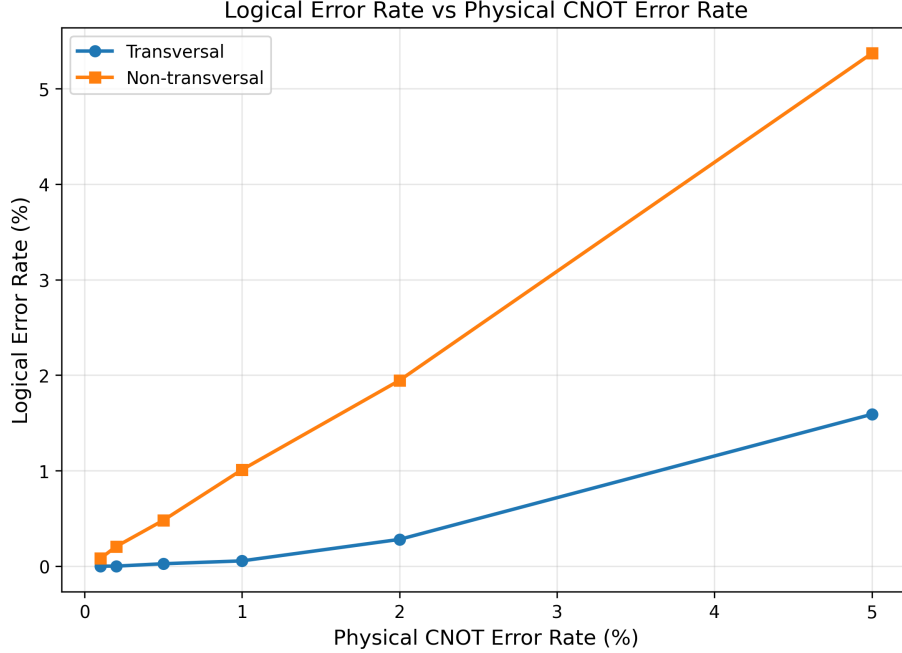


Figure 2: Logical error rate versus physical CNOT error rate for transversal and non-transversal logical CNOT constructions. The transversal construction suppresses first-order fault propagation and approaches quadratic scaling, whereas the non-transversal construction remains approximately linear.

5 Noise Sensitivity of the CIFAR-10 Quantum Classifier

5.1 Custom GPU statevector simulation

To evaluate the trained classifier under noise without the cost of density-matrix simulation, we implemented a custom eight-qubit statevector simulator in PyTorch and executed the stochastic trajectories on an NVIDIA RTX 4090 GPU. The statevector contains only $2^8 = 256$ complex amplitudes per trajectory, which makes batched Monte Carlo simulation practical over the full 10,000-image CIFAR-10 test set. The implementation reproduces the original noiseless test accuracy of 91.43%, providing a direct consistency check between the trained quantum model and the custom simulator.

For the first noise study, each modeled fault location is followed by an independent bit-flip channel. The four data-reuploading blocks contain data-encoding rotations, trainable R_Y and R_Z rotations, and an eight-gate CNOT ring. Counting the two CNOT participants separately gives approximately

$$4(8 + 8 + 8 + 16) = 160 \quad (3)$$

modeled single-qubit fault opportunities per inference. This depth is important because a small per-location error probability can accumulate into a substantial end-to-end logical disturbance.

5.2 Unprotected classification under bit-flip noise

Figure 3 shows the resulting accuracy degradation. At a physical bit-flip probability of only $p = 0.005$, the classifier still achieves approximately 88.87% accuracy, but at $p = 0.01$ it falls to

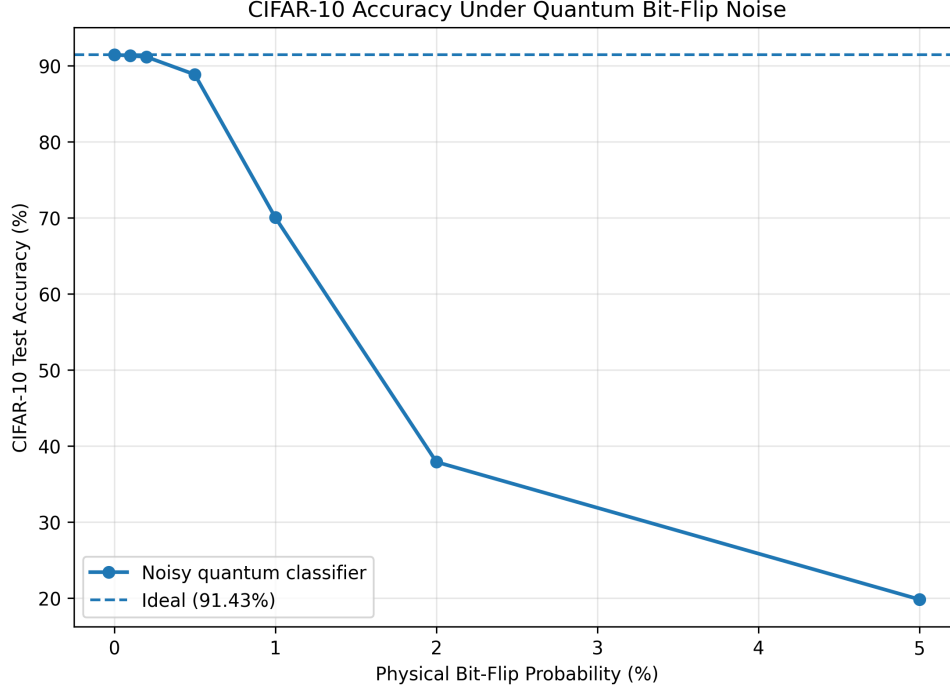


Figure 3: CIFAR-10 test accuracy of the unprotected eight-qubit classifier under stochastic bit-flip noise. The dashed line denotes the ideal 91.43% accuracy. The rapid decline reflects the accumulation of faults across approximately 160 modeled locations per inference.

70.75%. Increasing the error probability to $p = 0.02$ reduces the accuracy to 37.57%, and at $p = 0.05$ the model reaches only 19.75%. Thus, the classifier is highly sensitive to repeated quantum faults even when each individual error probability appears modest.

The same trend can be interpreted probabilistically. If one crudely approximates the locations as independent and equally harmful, the probability of experiencing at least one fault grows as

$$P_{\geq 1} = 1 - (1 - p)^L, \quad (4)$$

where $L \approx 160$. For example, even $p = 0.01$ gives $1 - (0.99)^{160} \approx 0.80$. Although not every physical fault produces an equally damaging logical effect, this estimate explains why end-to-end accuracy can degrade much faster than the nominal per-gate error rate might suggest. The classifier therefore provides a useful workload for studying whether error correction reduces the *accumulated* logical channel sufficiently to preserve a downstream task.

6 Ideal and Circuit-Level Repetition-Code Protection

6.1 Ideal repetition-code proxy

We next connected the three-qubit repetition-code result in Eq. (2) to the CIFAR-10 classifier. Instead of simulating three physical qubits for every logical qubit, the ideal-QEC proxy maps the physical bit-flip rate p to the exact repetition-code logical rate

$$p_L = 3p^2 - 2p^3, \quad (5)$$

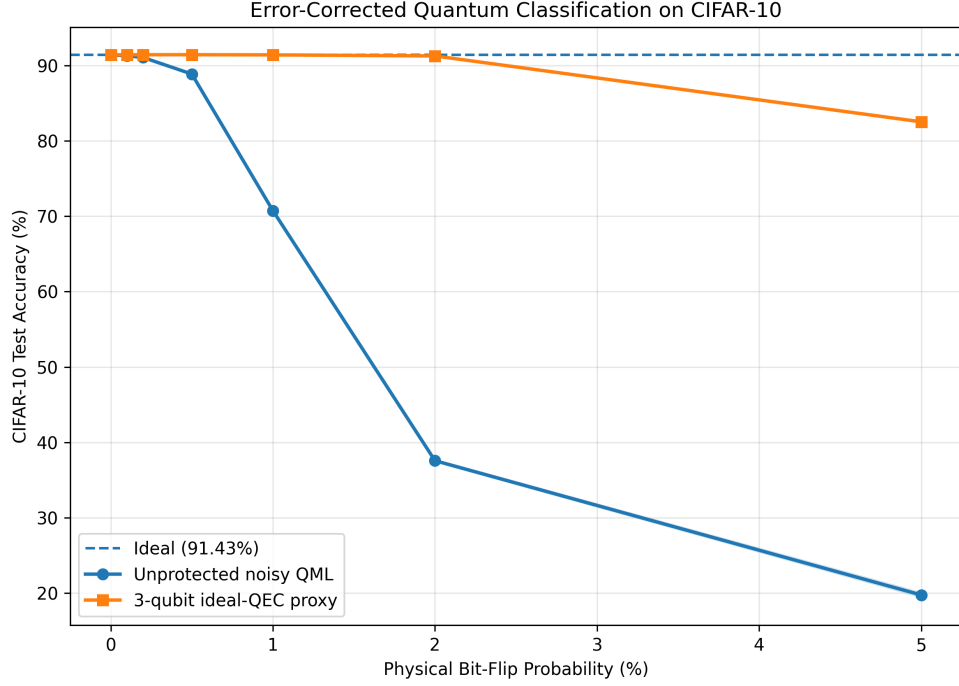


Figure 4: CIFAR-10 accuracy under unprotected bit-flip noise and an ideal three-qubit repetition-code proxy. Ideal error correction strongly suppresses the effective logical channel and preserves the task accuracy over the low- and moderate-noise regime.

and applies the resulting logical channel to the eight-qubit classifier. This isolates the benefit of perfect encoding, syndrome extraction, decoding, and recovery while leaving the trained classifier unchanged.

Figure 4 shows a large separation between the unprotected and ideal-QEC curves. The protected classifier remains essentially at its noiseless operating point through $p = 0.02$: at $p = 0.01$ the accuracy is approximately 91.41%, compared with 70.75% without protection, and at $p = 0.02$ it remains approximately 91.27% while the unprotected classifier falls to 37.57%. Even at $p = 0.05$, the ideal repetition-code proxy retains 82.53% accuracy compared with 19.75% unprotected.

The idealized logical channel also exhibits the expected distance-3 behavior. Because the leading term is $3p^2$, the physical-to-logical error mapping is approximately quadratic for small p . This experiment therefore establishes an optimistic upper bound: if the error-correction procedure were free and perfect, QEC would recover most of the task performance lost to physical noise.

6.2 Noisy syndrome measurements

The next model keeps the repetition-code recovery ideal but introduces syndrome-bit errors. With $p_{\text{syn}} = 0.01$, we compare one syndrome round against three repeated rounds followed by bit-wise majority voting. As shown in Fig. 5, both protected models remain near the ideal classifier for low physical data-error rates, while repeated syndrome extraction becomes increasingly useful at larger p_{data} . At $p_{\text{data}} = 0.05$, one-round noisy QEC achieves 75.54% accuracy, whereas three-round syndrome voting reaches 82.23%, an improvement of 6.69 percentage points. This experiment confirms that measurement repetition can improve task-level robustness when the principal uncertainty is the syndrome readout and the underlying data error is effectively fixed during repetition.

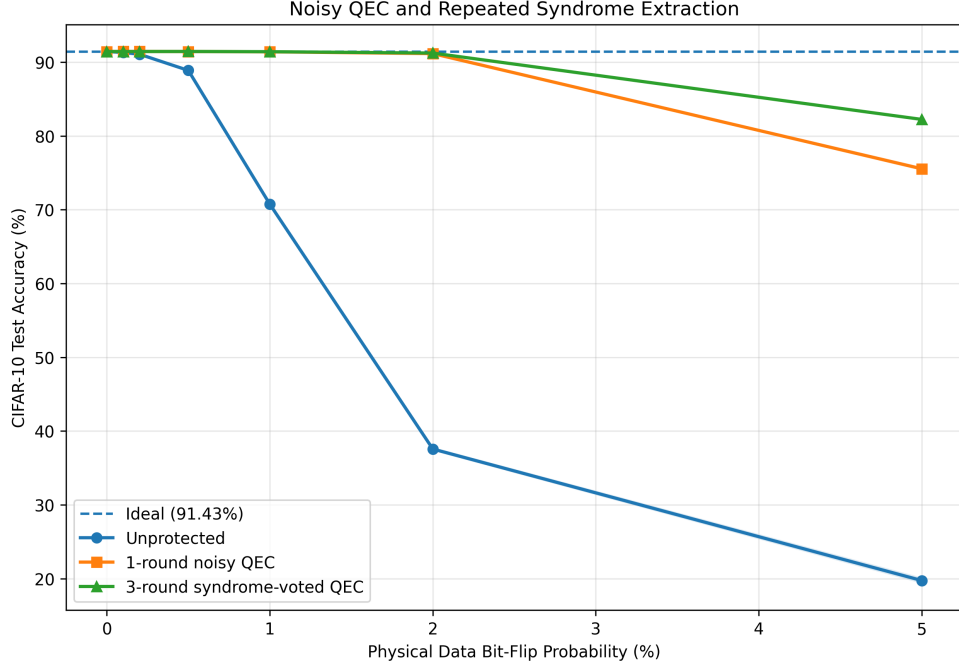


Figure 5: CIFAR-10 accuracy with noisy syndrome measurements. Repeating the syndrome three times and majority-voting the result improves robustness when syndrome noise dominates and additional rounds do not themselves introduce data faults.

6.3 Circuit-level QEC and the error-correction floor

The previous experiment still assumes that the extraction circuit itself is harmless. We therefore introduce a circuit-level repetition-code model with $p_{\text{CNOT}} = 0.005$, $p_{\text{meas}} = 0.01$, and $p_{\text{recovery}} = 0.005$. These errors produce a nonzero logical-error floor even when the external data-error probability is zero. In the corresponding simulation, the effective logical error rates at $p_{\text{data}} = 0$ are approximately 0.0051 for one round and 0.0062 for three rounds.

This floor is immediately visible at the task level. As shown in Fig. 6, the unprotected classifier remains at 91.43% when $p_{\text{data}} = 0$, but one-round QEC reaches only 88.40% and three-round QEC only 85.91%. Thus, when the data are already clean, the correction circuit does more harm than good. As the physical data-noise rate increases, however, QEC eventually becomes beneficial. At $p_{\text{data}} = 0.01$, the unprotected classifier falls to 70.75%, while one- and three-round circuit-level QEC achieve approximately 86.25% and 84.12%, respectively. At $p_{\text{data}} = 0.05$, both QEC variants remain near 50%, compared with 19.75% without protection.

These results expose the central engineering tradeoff of fault tolerance:

$$\text{net benefit of QEC} = \text{errors removed} - \text{errors introduced by the QEC gadget}. \quad (6)$$

Increasing the number of syndrome rounds improves confidence in the measured syndrome, but also adds more faulty gates and measurements. Consequently, three rounds are not uniformly superior to one round.

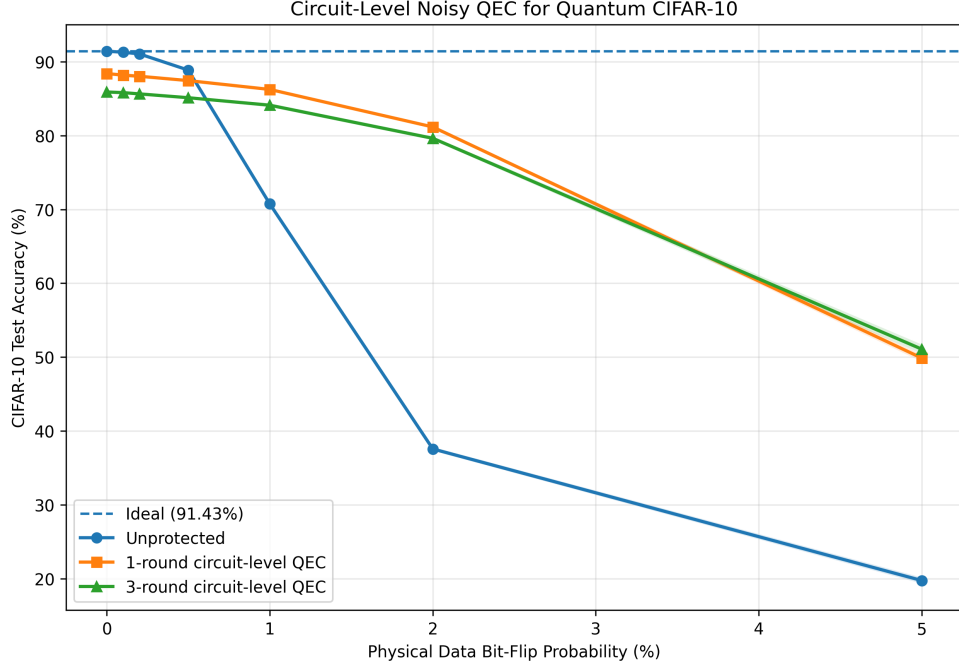


Figure 6: Circuit-level repetition-code QEC with noisy syndrome CNOTs, measurements, and recovery. The QEC circuitry introduces a nonzero error floor, so protection is harmful at very low data noise but becomes beneficial after a break-even region is crossed.

6.4 Break-even behavior for repeated syndrome extraction

To quantify this tradeoff, we swept the syndrome-CNOT error rate and measurement error rate for several physical data-noise levels. For each parameter pair we computed the task-level gain

$$\Delta A = A_{\text{QEC3}} - A_{\text{QEC1}}. \quad (7)$$

At $p_{\text{data}} = 0.005$ and 0.01 , QEC-3 is better than QEC-1 for 43.3% of the tested parameter combinations; this fraction rises to 46.7% at $p_{\text{data}} = 0.02$ and 60.0% at $p_{\text{data}} = 0.05$. The heatmap in Fig. 7 illustrates the $p_{\text{data}} = 0.05$ case. When the measurement error is 5% and the syndrome CNOT is ideal, three rounds improve classification accuracy by approximately 28.1 percentage points. In contrast, at the same measurement error but with a 2% syndrome-CNOT error rate, the gain becomes approximately -1.8 percentage points. The zero contour therefore marks an empirical break-even boundary between a measurement-noise-dominated regime, where repetition helps, and a gate-noise-dominated regime, where additional extraction is harmful.

The repetition-code studies therefore establish three progressively stronger conclusions. First, ideal distance-3 QEC can strongly suppress accumulated classifier noise. Second, repeated syndrome measurements are beneficial when measurement uncertainty dominates. Third, once the extraction circuit itself is noisy, the optimal number of QEC rounds becomes operating-point dependent. These observations motivate the transition to a code capable of correcting general Pauli errors and to stabilizer-extraction circuits designed explicitly to control fault propagation.

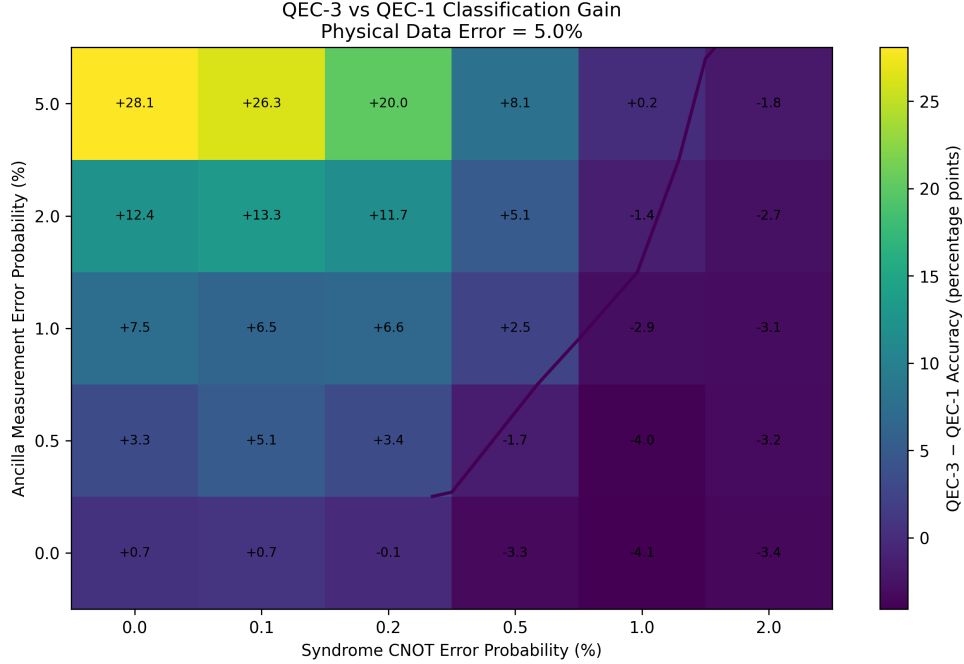


Figure 7: Task-level gain of three-round over one-round circuit-level repetition-code QEC at $p_{\text{data}} = 0.05$. Positive values indicate that repeated syndrome extraction improves CIFAR-10 accuracy. The contour near zero separates the region in which additional syndrome information is beneficial from the region in which added faulty circuitry dominates.

7 General Pauli Noise and the $[[5, 1, 3]]$ Perfect Code

Motivation. The three-qubit repetition-code experiments provided a transparent setting for understanding error detection, syndrome extraction, recovery, and fault propagation; however, the repetition code protects only against bit-flip errors. A more realistic quantum-noise model must account for all three single-qubit Pauli errors, X , Y , and Z . We therefore replaced the bit-flip channel with the single-qubit depolarizing channel

$$\mathcal{E}_p(\rho) = (1 - p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z), \quad (8)$$

where p denotes the total probability of a non-identity Pauli error. To protect against this more general error model, we adopted the $[[5, 1, 3]]$ perfect quantum error-correcting code, which encodes one logical qubit into five physical qubits and corrects an arbitrary single-qubit Pauli error.

Five-qubit stabilizer code. The code was represented using the four stabilizer generators

$$S_1 = XZZXI, \quad S_2 = IXZZX, \quad S_3 = XIXZZ, \quad S_4 = ZXIXZ, \quad (9)$$

with logical operators

$$\bar{X} = XXXXX, \quad \bar{Z} = ZZZZZ. \quad (10)$$

Because the code has distance $d = 3$, every weight-one Pauli error produces a distinguishable syndrome and can be corrected. In the present CIFAR-10 classifier, eight logical qubits are used; therefore, an explicitly encoded representation would require at least

$$8 \times 5 = 40 \quad (11)$$

Table 2: Exact logical error probability of the ideal $[[5, 1, 3]]$ code under independent single-qubit depolarizing noise.

Physical error p	Logical error p_L
0.000	0.000000
0.001	0.000010
0.002	0.000040
0.005	0.000247
0.010	0.000978
0.020	0.003825
0.050	0.022332

physical data qubits before including syndrome and flag ancillas. Simulating a full 2^{40} -dimensional statevector would be prohibitively expensive for repeated evaluation over 10,000 CIFAR-10 test samples. We therefore retained the eight-qubit logical statevector while treating each five-qubit encoded block through Pauli-frame and stabilizer calculations.

Exact logical-channel construction. For the ideal-code experiment, all

$$4^5 = 1024 \tag{12}$$

five-qubit Pauli patterns were enumerated exactly. For each physical error pattern, the four-bit stabilizer syndrome was computed, the corresponding minimum-weight correction was applied, and the residual operator was classified as one of

$$I_L, \quad X_L, \quad Y_L, \quad Z_L. \tag{13}$$

This produced an exact effective logical Pauli channel for each physical depolarizing probability p . The resulting total logical error probability was

$$p_L = 1 - P(I_L). \tag{14}$$

Table 2 reports the resulting physical-to-logical error mapping.

Logical error suppression. Figure 8 compares the physical depolarizing probability with the resulting logical error probability. In the low-noise regime, the logical error is substantially smaller than the physical error. For example, at $p = 0.01$, the physical non-identity error probability is 1%, whereas the ideal five-qubit code reduces the logical error probability to approximately

$$p_L = 9.78 \times 10^{-4}. \tag{15}$$

At $p = 0.02$, the corresponding logical error remains only 3.83×10^{-3} . This suppression reflects the distance-three property of the code: all first-order single-qubit Pauli faults are correctable, so the dominant uncorrectable contribution arises from higher-weight events.

Effect on CIFAR-10 classification. The logical Pauli channel obtained above was then inserted into the eight-logical-qubit CIFAR-10 classifier. Table 3 compares the unprotected classifier with the ideal $[[5, 1, 3]]$ -protected model. The ideal classifier accuracy was 91.43%. At low and moderate physical error rates, the encoded model remained close to this ideal value. At $p = 0.01$, for example,

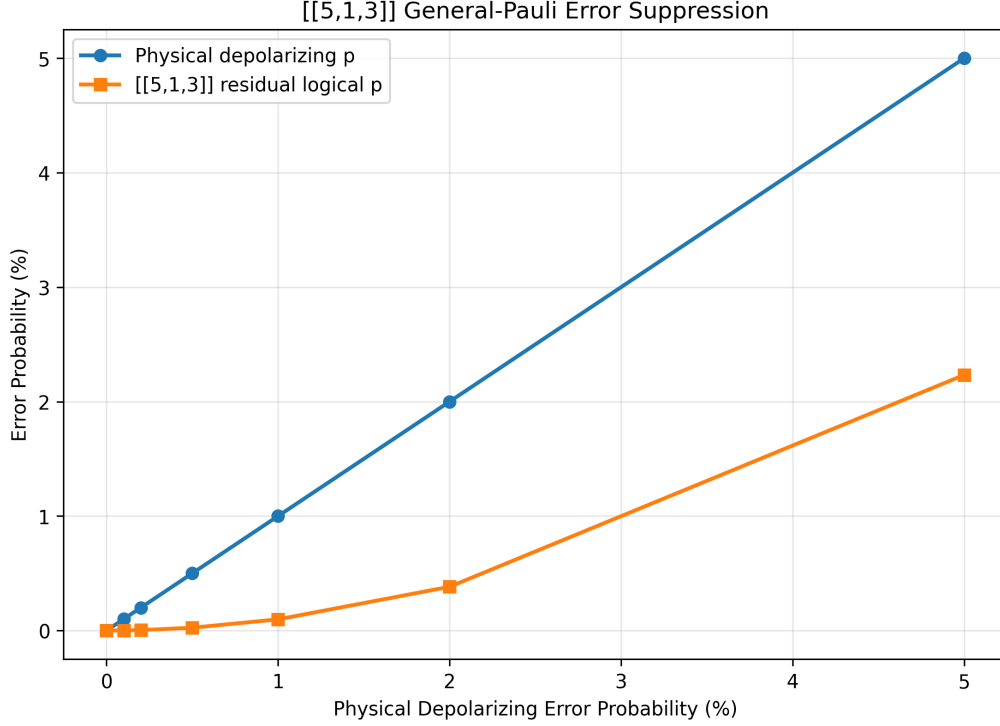


Figure 8: Physical depolarizing error probability and the exact residual logical error probability of the ideal $[[5, 1, 3]]$ code. The code strongly suppresses logical faults in the low-noise regime.

the unprotected classifier decreased to 74.74%, whereas ideal five-qubit protection preserved 91.30% accuracy. At $p = 0.02$, the separation became even larger: the unprotected accuracy dropped to 34.43%, while the protected classifier retained 90.64%. These results demonstrate the potential benefit of QEC when the encoding, syndrome extraction, and recovery operations themselves are assumed ideal.

Accuracy recovery. Figure 10 shows the classification accuracy recovered by ideal five-qubit QEC relative to the unprotected classifier. The largest gains occur once accumulated circuit noise has already begun to substantially degrade the unprotected VQC. The result establishes an important reference point for the remainder of this study: the code itself is capable of strong error suppression, but this result represents an optimistic upper bound because the correction circuitry has not yet been allowed to fail.

Residual logical Pauli channel. In addition to the total logical error probability, the simulation retained the residual X_L , Y_L , and Z_L probabilities. Under the symmetric physical depolarizing model and the five-qubit code, the residual logical components remained approximately symmetric, as shown in Fig. 11. This complete logical-channel representation was subsequently used instead of reducing every fault to a bit-flip-only model.

Key observation. The ideal-code experiment demonstrates that the poor performance of a noisy quantum classifier is not fundamentally unavoidable: if physical Pauli errors can be converted into sufficiently small logical error probabilities, the original 91.43% classifier can be preserved over a

Table 3: CIFAR-10 accuracy under physical depolarizing noise with and without ideal $[[5, 1, 3]]$ protection.

p	Unprotected	Ideal $[[5, 1, 3]]$
0.000	0.9143	0.9143
0.001	0.9133	0.9143
0.002	0.9117	0.9142
0.005	0.8974	0.9141
0.010	0.7474	0.9130
0.020	0.3443	0.9064
0.050	0.1319	0.2919

much wider noise range. However, this experiment does not yet address the central FTQC problem: the stabilizer measurements, ancilla preparation, entangling gates, measurements, and recovery operations required to perform QEC are themselves noisy. The following section explicitly introduces those faults.

8 Noisy Stabilizer Extraction and Flagged Fault-Tolerant Error Correction

Circuit-level QEC noise. We next replaced ideal syndrome extraction with an explicit noisy stabilizer-measurement model. The five physical data qubits continued to experience general X , Y , and Z errors, while the error-correction circuitry was assigned the fixed noise parameters

$$p_{2Q} = 0.005, \quad p_{1Q} = 0.001, \quad p_{\text{prep}} = 0.001, \quad p_{\text{meas}} = 0.01, \quad p_{\text{rec}} = 0.005. \quad (16)$$

Here, p_{2Q} denotes two-qubit depolarizing noise following syndrome-extraction CNOTs, p_{1Q} denotes one-qubit depolarizing noise following basis-changing Hadamard gates, p_{prep} models ancilla-preparation error, p_{meas} is the classical measurement-bit error probability, and p_{rec} models a physical fault following the recovery operation. The simulator propagated these faults explicitly in the Pauli frame of each encoded block.

Naive single-ancilla stabilizer extraction. In the first circuit-level implementation, each weight-four stabilizer was measured with a single syndrome ancilla. For a stabilizer containing an X component, the corresponding data qubit was basis-rotated with a Hadamard gate, coupled to the syndrome ancilla through a CNOT, and rotated back. The same syndrome ancilla interacted sequentially with all four supported data qubits. Although this circuit correctly measures the stabilizer in the absence of faults, it is not fault tolerant. In particular, a Pauli fault on the syndrome ancilla can propagate backward through subsequent CNOTs and create a correlated error on multiple data qubits. For a distance-three code, a single physical fault that produces a weight-two data error can already exceed the guaranteed correction capability of the code.

Large QEC-induced logical-error floor. The effect of this fault propagation was severe. Even when the externally applied physical data-error probability was set to

$$p = 0, \quad (17)$$

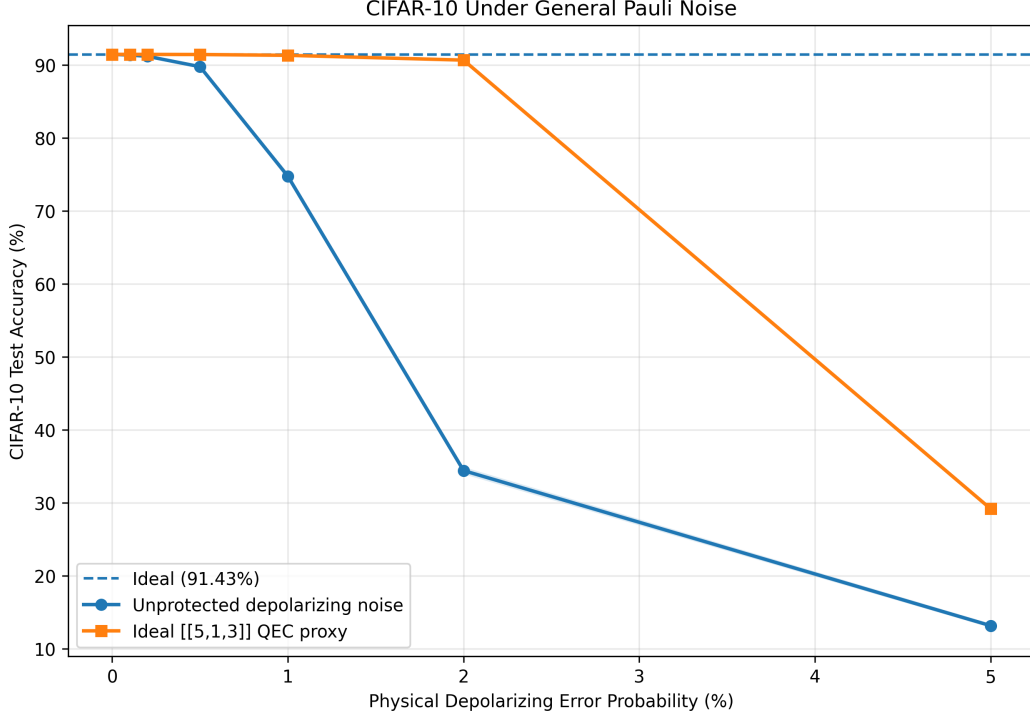


Figure 9: CIFAR-10 classification accuracy under general depolarizing noise. Ideal $[[5, 1, 3]]$ protection preserves the trained classifier accuracy over a substantially larger noise range than the unprotected logical circuit.

the noisy correction gadget produced

$$p_L^{\text{naive}} = 0.039094. \quad (18)$$

Thus, the QEC circuit itself generated a logical-error probability of approximately 3.91% per modeled protected location before any swept data noise was introduced. When this effective logical channel was propagated through the full classifier, CIFAR-10 accuracy collapsed from the ideal 91.43% to approximately

$$15.10\%. \quad (19)$$

This experiment provides a direct numerical example of the distinction between quantum error correction and fault-tolerant quantum error correction: a code may correct arbitrary single-qubit Pauli errors ideally while a non-fault-tolerant syndrome-extraction circuit can convert one circuit fault into an uncorrectable correlated error.

Flagged stabilizer extraction. To suppress such hook-error propagation, we introduced a second ancilla—the *flag ancilla*—during stabilizer extraction. The syndrome ancilla continued to record the stabilizer parity, while the flag ancilla interacted with the syndrome ancilla at selected points in the extraction sequence. A flag event indicated that a fault may have propagated into a dangerous correlated data error. When either a flag was raised or the measured stabilizer syndrome was nontrivial, the decoder performed a complete four-stabilizer follow-up extraction and selected a correction using a flag-aware lookup table.

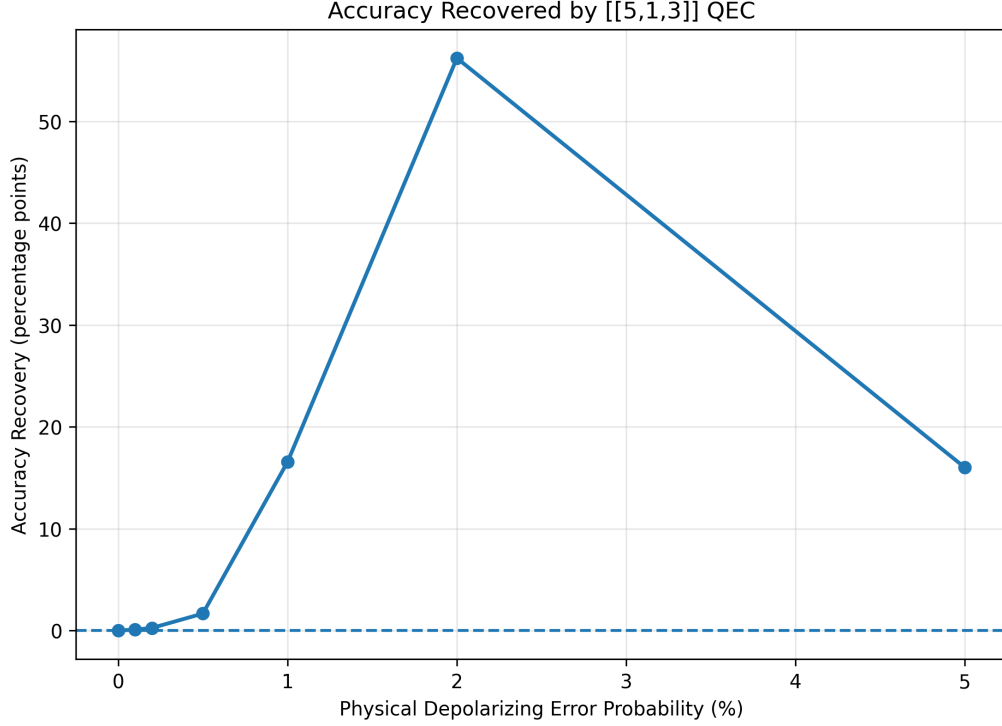


Figure 10: CIFAR-10 accuracy recovered by ideal $[[5,1,3]]$ QEC relative to the corresponding unprotected classifier.

Flag-aware correlated-error decoding. Rather than assuming that every nontrivial syndrome corresponds to a single-qubit error, the flagged decoder explicitly included correlated errors generated by single faults in the flagged extraction circuit. For the first stabilizer, $XZZXI$, the implementation verified eight distinct flag-conditioned single-fault error classes,

$$\{IIIII, IIZXI, IXZXI, IYZXI, IZZXI, IIIXI, IIXXI, IIYXI\}. \quad (20)$$

Equivalent automatically generated lookup tables were used for the remaining three stabilizers. This allowed the decoder to use both the measured syndrome and flag information to distinguish dangerous correlated errors that would be ambiguous to the ordinary minimum-weight decoder.

Logical-error reduction. The flagged procedure produced a substantial reduction in the QEC-induced logical-error rate. Table 4 compares the naive and flagged correction gadgets. At $p = 0$, the logical error decreased from

$$0.039094 \quad \text{to} \quad 0.013274, \quad (21)$$

corresponding to a relative reduction of approximately

$$\frac{0.039094 - 0.013274}{0.039094} \times 100\% \approx 66.0\%. \quad (22)$$

The flagged advantage persisted throughout the tested physical-noise range. At $p = 0.01$, for example, the logical error decreased from 0.042086 to 0.019848, while at $p = 0.02$ it decreased from 0.046420 to 0.027324.

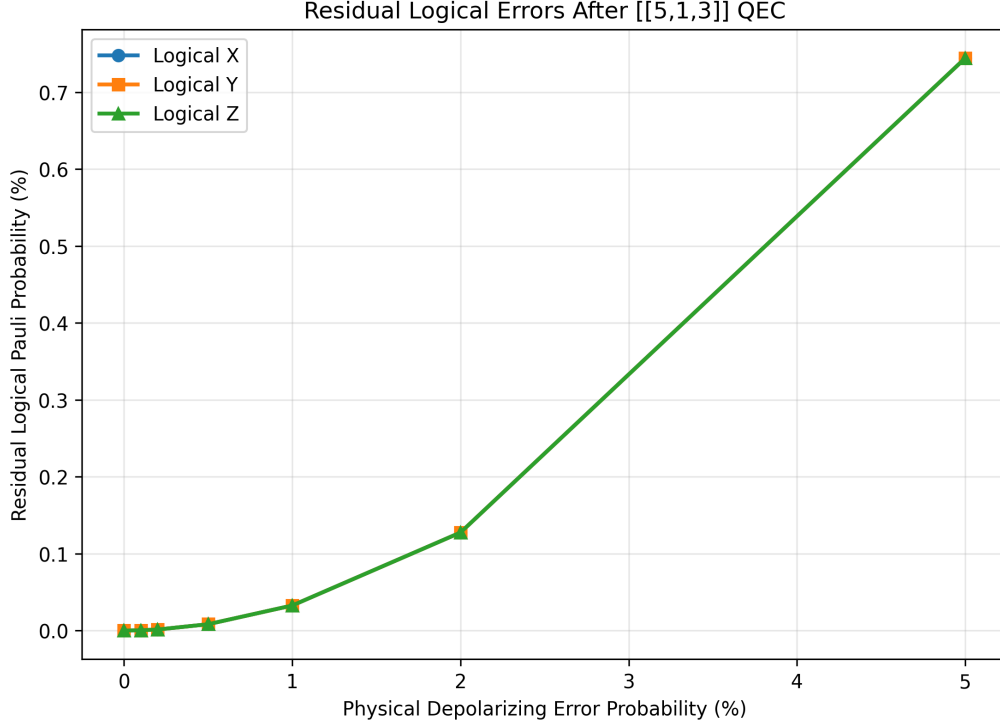


Figure 11: Residual logical Pauli distribution of the ideal $[[5, 1, 3]]$ code under physical depolarizing noise.

End-task classification improvement. The improvement in logical error translated directly to CIFAR-10 performance. Table 5 reports the corresponding classification accuracies. At $p = 0$, naive noisy stabilizer extraction achieved only 15.10%, whereas flagged FTEC increased the accuracy to

$$59.21\%. \quad (23)$$

At $p = 0.01$, the improvement was from 14.19% to 35.20%, and at $p = 0.02$ the accuracy increased from 13.78% to 22.33%. The difference narrowed at $p = 0.05$, where both circuits operated close to the low-accuracy regime.

Adaptive branch behavior. The flagged protocol does not perform the expensive follow-up path on every correction cycle. Figure 14 reports how frequently the flag branch, nontrivial-syndrome branch, and no-trigger branch were selected. At $p = 0$, approximately 82.16% of correction cycles required no additional branch, while 7.88% triggered the flag path and 9.96% triggered the nontrivial-syndrome path. As physical data noise increased, the nontrivial-syndrome branch became increasingly common, reaching 29.28% at $p = 0.05$. This adaptive behavior is important because it demonstrates that the additional diagnostic circuitry is invoked only when the measured fault information indicates that it is needed.

Flagged-FTEC gain. Figure 15 isolates the difference between flagged and naive circuit-level QEC at the classification level. The large positive gain at low and moderate physical noise confirms that the dominant failure mechanism in the naive circuit is not simply the presence of gate noise, but the manner in which individual faults propagate through the extraction circuit. Introducing a

Table 4: Residual logical error probability of naive noisy stabilizer extraction and flagged FTEC for the $[[5, 1, 3]]$ code.

p	Ideal p_L	Naive QEC p_L	Flagged FTEC p_L
0.000	0.000000	0.039094	0.013274
0.001	0.000010	0.039524	0.014274
0.002	0.000040	0.039228	0.014294
0.005	0.000247	0.040244	0.016382
0.010	0.000978	0.042086	0.019848
0.020	0.003825	0.046420	0.027324
0.050	0.022332	0.070316	0.056772

Table 5: CIFAR-10 accuracy obtained using naive noisy stabilizer QEC and flagged FTEC. Mean values are reported over five stochastic classifier seeds.

p	Unprotected	Ideal $[[5, 1, 3]]$	Naive QEC	Flagged FTEC
0.000	0.9143	0.9143	0.1510	0.5921
0.001	0.9133	0.9143	0.1484	0.5416
0.002	0.9117	0.9142	0.1511	0.5495
0.005	0.8974	0.9141	0.1473	0.4553
0.010	0.7474	0.9130	0.1419	0.3520
0.020	0.3443	0.9064	0.1378	0.2233
0.050	0.1319	0.2919	0.1204	0.1251

flag ancilla does not eliminate circuit noise; instead, it provides additional information that allows dangerous propagated faults to be identified and decoded more appropriately.

Remaining error floor. Despite the large improvement, flagged FTEC still produced

$$p_L = 0.013274 \quad (24)$$

even at $p = 0$. Therefore, the correction gadget remained far from the ideal five-qubit-code behavior. This residual floor originates from faults in syndrome CNOTs, basis rotations, ancilla preparation, measurements, follow-up extraction, and recovery. Because the classifier contains many logical operation locations, even a logical error probability near 1% can accumulate into several logical faults during one inference pass. Thus, flagged extraction solves an important fault-propagation problem, but it does not by itself guarantee that the complete logical algorithm is fault tolerant.

Transition to temporal decoding. The flagged experiment also motivates a second issue. If stabilizers are measured repeatedly, new physical faults can occur between measurement rounds. Therefore, the measured syndrome is generally time dependent:

$$s^{(1)} \neq s^{(2)} \neq s^{(3)}. \quad (25)$$

A bitwise majority vote over the raw syndromes may incorrectly interpret a genuine data fault that appeared between rounds as a measurement error. The next section therefore evaluates whether the temporal changes

$$\Delta s_t = s^{(t)} \oplus s^{(t-1)} \quad (26)$$

provide more useful decoding information than raw repeated-syndrome majority voting.

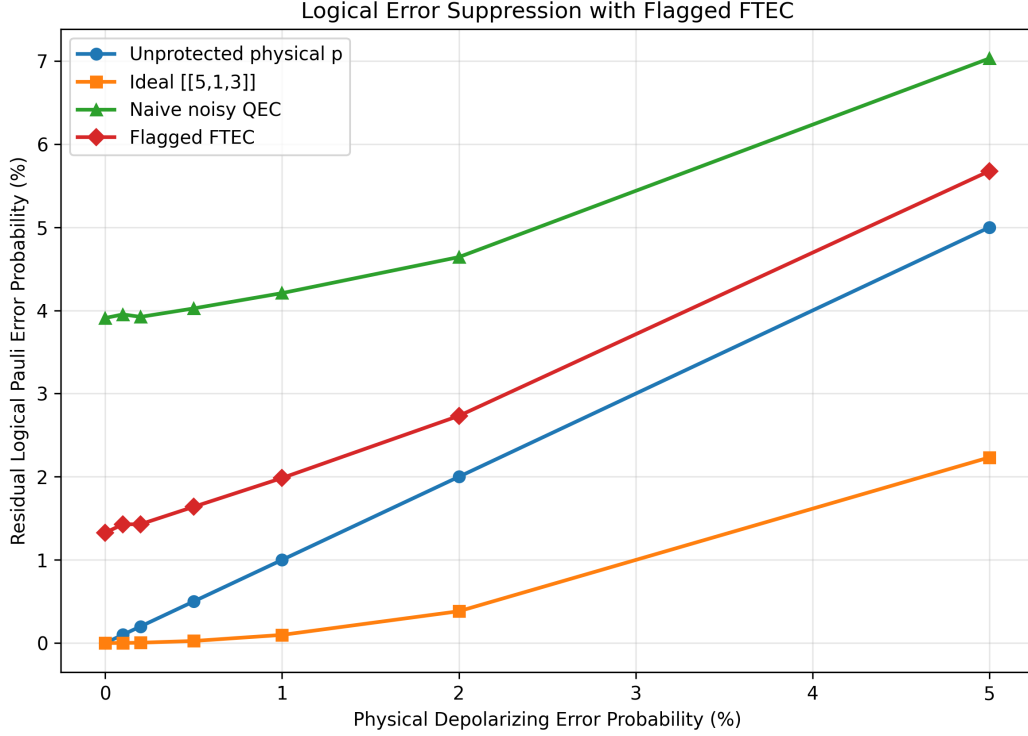


Figure 12: Residual logical error probability for ideal five-qubit QEC, naive noisy stabilizer extraction, and flagged FTEC. Flag information substantially reduces the logical-error floor caused by noisy extraction circuitry.

9 Syndrome-History Decoding

Motivation. The flagged-FTEC experiment substantially reduced correlated hook errors, but it also highlighted a limitation of repeated syndrome extraction. When stabilizers are measured several times, the physical data state is not necessarily unchanged between rounds because every extraction round contains noisy gates, ancilla preparation, and measurement. Consequently, a sequence of measured syndromes

$$s^{(1)}, \quad s^{(2)}, \quad s^{(3)} \quad (27)$$

cannot always be interpreted as repeated noisy observations of one fixed underlying syndrome. A new data fault may occur between rounds, causing a genuine temporal change in the syndrome. This makes simple bitwise majority voting potentially suboptimal.

Temporal detection events. To retain information about when syndrome changes occurred, we constructed detection events from three complete flagged syndrome-extraction rounds. For the first round, the observed syndrome itself was used,

$$d^{(1)} = s^{(1)}, \quad (28)$$

while subsequent detection events were defined as

$$d^{(2)} = s^{(2)} \oplus s^{(1)}, \quad d^{(3)} = s^{(3)} \oplus s^{(2)}. \quad (29)$$

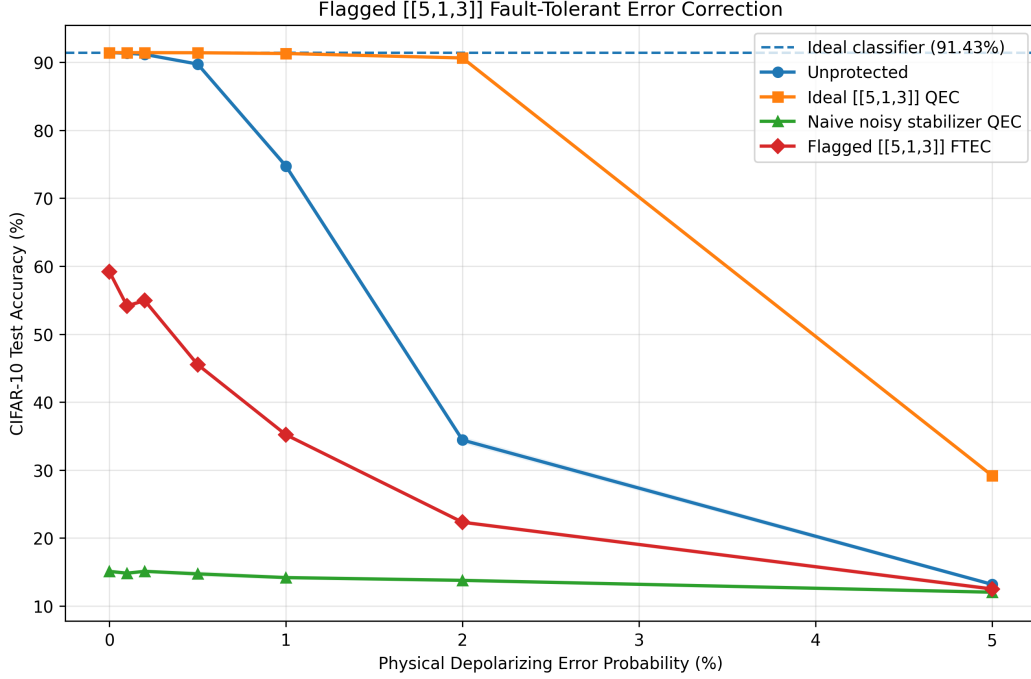


Figure 13: CIFAR-10 accuracy obtained with unprotected inference, ideal $[[5, 1, 3]]$ protection, naive noisy stabilizer extraction, and flagged FTEC. Flagged extraction recovers a substantial portion of the accuracy lost to non-fault-tolerant syndrome circuitry.

Each syndrome contains four stabilizer bits, producing 12 temporal detection-event bits. In addition, four flag-summary bits were recorded, where the flag-summary bit for stabilizer j was set when that stabilizer raised a flag in any of the three rounds. The resulting decoder input was therefore a 16-bit history vector,

$$h = [d^{(1)}, d^{(2)}, d^{(3)}, f_{\text{any}}]. \quad (30)$$

Maximum-likelihood lookup decoder. Because the $[[5, 1, 3]]$ code has a very small state space, a direct lookup decoder was used instead of a large-code graph decoder. The final physical Pauli frame was represented by one of

$$16 \times 4 = 64 \quad (31)$$

coset classes, corresponding to 16 possible final stabilizer syndromes and four residual logical classes,

$$\{I_L, X_L, Y_L, Z_L\}. \quad (32)$$

For every observed 16-bit history key, the decoder selected the most frequently observed final coset in a synthetic calibration set. The decoder was calibrated using 100,000 circuit-level samples for each physical-noise value in the sweep, giving a total of 700,000 calibration histories. Importantly, these samples contained only the quantum circuit-noise model; CIFAR-10 labels were never used to train the decoder. Decoder performance was subsequently evaluated on an independent set of 300,000 encoded-block simulations per physical-noise value.

Comparison with raw majority voting. Three decoder strategies were compared: the previously developed adaptive flagged-FTEC procedure, a three-round bitwise majority vote over the

Table 6: Branch frequencies of the flagged FTEC procedure.

p	Flag trigger	Syndrome trigger	No trigger
0.000	0.0788	0.0996	0.8216
0.001	0.0784	0.1041	0.8174
0.002	0.0783	0.1083	0.8135
0.005	0.0777	0.1219	0.8004
0.010	0.0764	0.1416	0.7820
0.020	0.0751	0.1829	0.7420
0.050	0.0696	0.2928	0.6376

Table 7: Residual logical error probability of the three-round syndrome-history decoder.

Physical error p	History-decoder p_L
0.000	0.053720
0.001	0.053777
0.002	0.054060
0.005	0.055977
0.010	0.060150
0.020	0.066053
0.050	0.096070

raw measured syndromes, and the three-round syndrome-history decoder. Figure 16 shows the resulting logical error probabilities. Temporal history consistently reduced the logical error relative to raw three-round majority voting. Across the complete physical-noise sweep, the mean logical error probability decreased from

$$\overline{p}_L^{\text{majority}} = 0.083861 \quad (33)$$

to

$$\overline{p}_L^{\text{history}} = 0.062830. \quad (34)$$

This confirms that the changes between consecutive syndromes contain useful information that is lost when the raw syndrome values are simply majority voted.

History-decoder logical error. Table 7 reports the logical error probability obtained from the syndrome-history decoder. Even at zero externally applied data noise, the three-round circuit produced

$$p_L^{\text{history}} = 0.053720, \quad (35)$$

because the syndrome-extraction circuitry itself remained noisy. The logical error increased gradually to 0.060150 at $p = 0.01$, 0.066053 at $p = 0.02$, and 0.096070 at $p = 0.05$.

Why temporal decoding did not outperform adaptive flagged FTEC. Although temporal decoding improved substantially over raw majority voting, it remained inferior to the previous adaptive flagged-FTEC strategy. The average logical error of adaptive flagged FTEC over the same physical-noise sweep was only

$$\overline{p}_L^{\text{flagged}} = 0.023167, \quad (36)$$

which was considerably smaller than both the history decoder and the raw-majority decoder. The reason is that the adaptive flagged procedure usually terminates without performing three complete

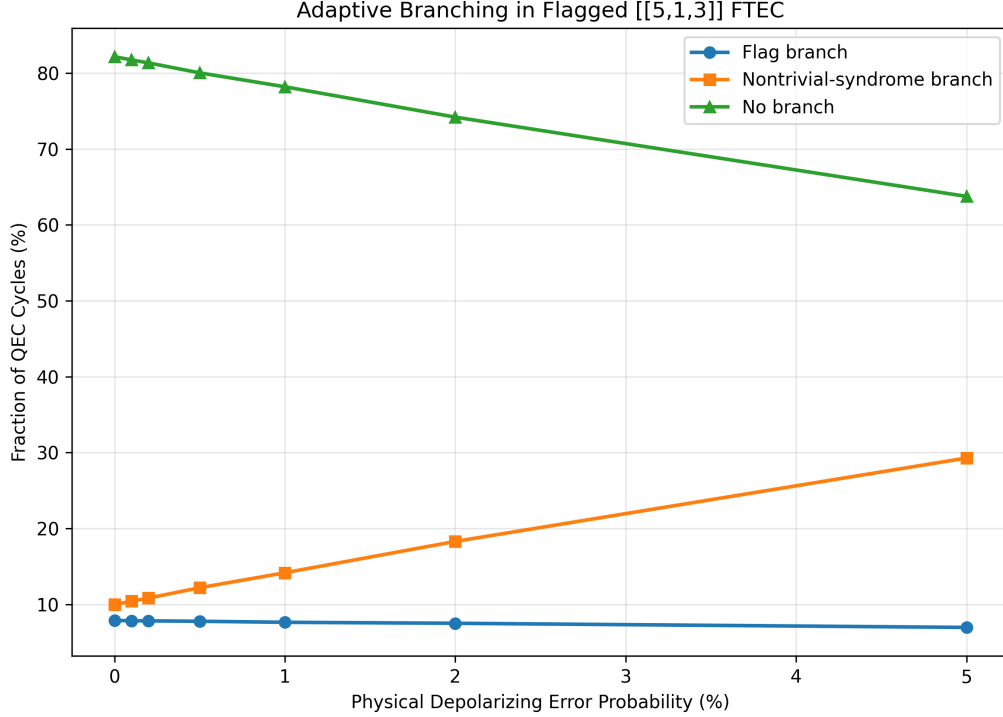


Figure 14: Adaptive branch rates of the flagged $[[5,1,3]]$ correction procedure. Most cycles require no follow-up at low physical noise, while syndrome-triggered follow-up becomes increasingly common as the physical data-error probability increases.

syndrome rounds, whereas the history decoder always incurs the noise associated with three full extractions. Thus, the history decoder obtains more information, but it obtains that information by executing substantially more noisy circuitry.

Classification-level consequence. Figure 17 shows the same tradeoff at the CIFAR-10 level. The previous adaptive flagged-FTEC method retained substantially higher classification accuracy than either of the three-round approaches. Raw majority voting and history decoding both operated close to the low-accuracy regime because their logical-error floors were applied repeatedly throughout the quantum classifier. Nevertheless, the history decoder generally produced a small improvement over raw majority voting at low and moderate physical error rates, demonstrating that its temporal information was useful even though the overall three-round strategy remained too noisy.

Value of temporal information. Figure 18 directly plots the classification difference

$$\Delta A_{\text{history}} = A_{\text{history}} - A_{\text{majority}}. \quad (37)$$

The history decoder provided a positive gain throughout most of the investigated low- and moderate-noise regime, with the improvement remaining below one percentage point. At the highest tested noise level, $p = 0.05$, the advantage disappeared and became slightly negative. This behavior is consistent with a regime in which the additional temporal information is useful when the syndrome history remains informative, but increasingly difficult to exploit once the extraction circuits themselves accumulate large numbers of physical faults.

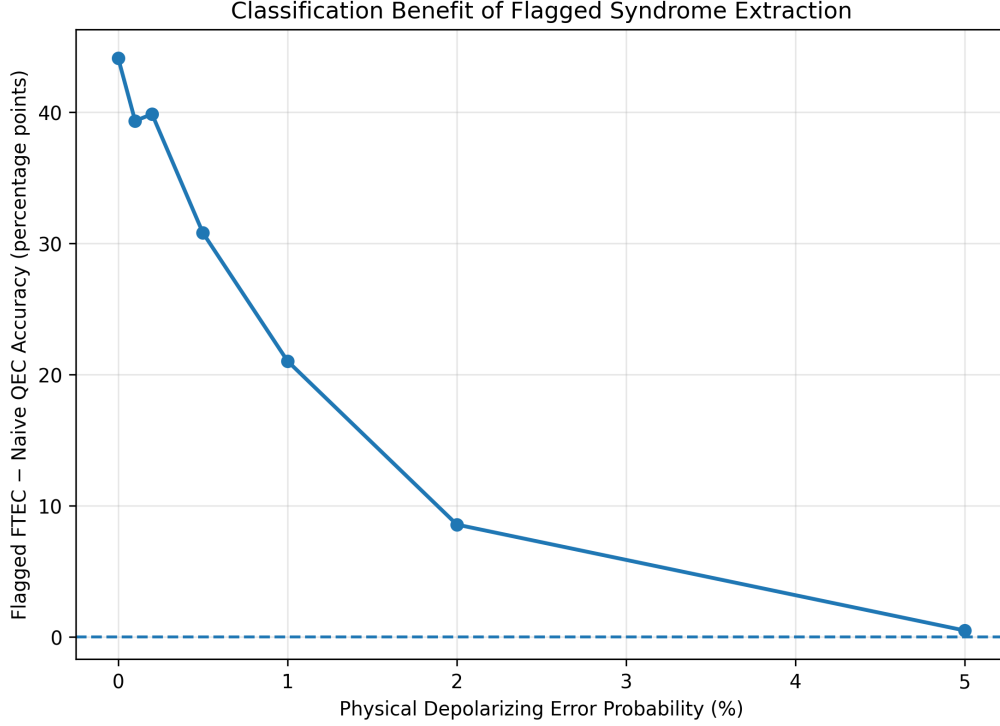


Figure 15: CIFAR-10 classification gain of flagged FTEC relative to naive noisy stabilizer extraction.

Coset-prediction accuracy. The maximum-likelihood decoder was also evaluated directly on its intended task: predicting the final physical Pauli coset from the observed syndrome and flag history. As shown in Fig. 19, the prediction accuracy was approximately 88% in the lowest-noise regime and decreased gradually as the physical error rate increased, reaching approximately 84% at $p = 0.05$. The relatively high coset-prediction accuracy confirms that the history carries meaningful information about the final physical error state. However, even an imperfectly predicted coset can result in a residual logical Pauli error, and the resulting 5–10% logical-error probabilities remain too high for repeated use throughout a deep classification circuit.

Temporal syndrome activity. Figure 20 provides further evidence that the syndrome is not static across the three extraction rounds. Approximately 36% of simulated correction cycles exhibited at least one syndrome change between consecutive rounds, whereas roughly 22–23% exhibited at least one flag event. The high temporal-change frequency persists even at very low externally applied data-noise probability because faults continue to arise from syndrome CNOTs, basis rotations, ancilla preparation, and measurement. Thus, treating repeated syndromes as independent noisy measurements of one fixed syndrome would discard relevant circuit-level information.

Decoder-selection result. The three decoder strategies were subsequently compared globally before constructing the final logical-gate model. Their mean logical error probabilities over the complete physical-noise sweep were

$$\overline{p}_L^{\text{flagged}} = 0.023167, \quad \overline{p}_L^{\text{history}} = 0.062830, \quad \overline{p}_L^{\text{majority}} = 0.083861. \quad (38)$$

The adaptive flagged-FTEC decoder therefore provided the smallest logical error and was selected as the fixed decoder for the subsequent end-to-end fault-tolerant classifier experiments.

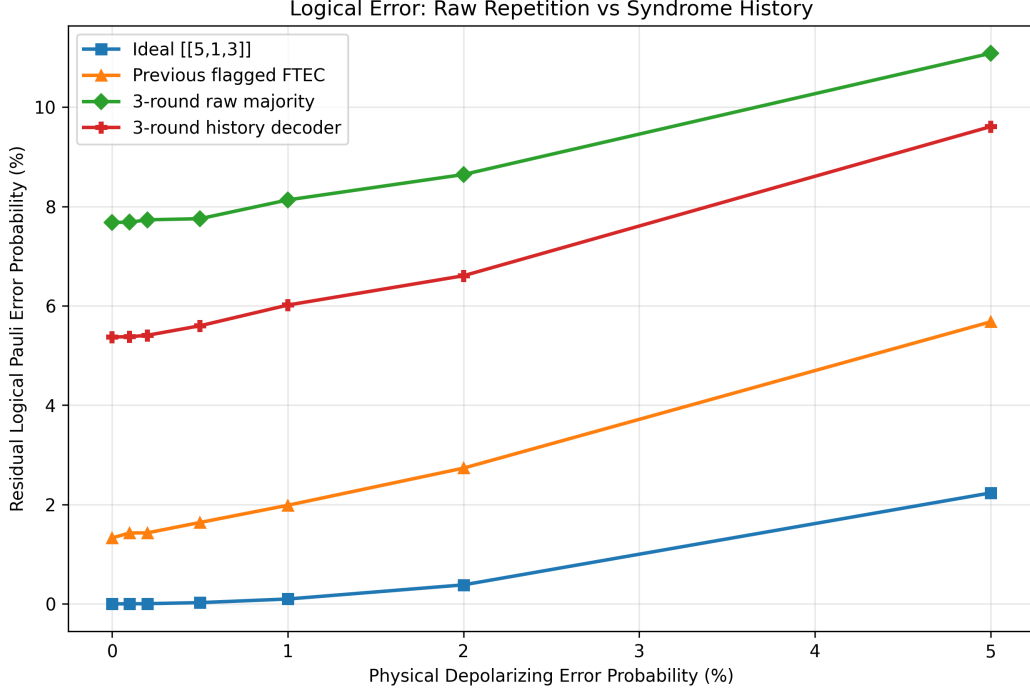


Figure 16: Residual logical Pauli error probability obtained using the ideal $[[5, 1, 3]]$ decoder, the previous adaptive flagged-FTEC procedure, three-round raw-syndrome majority voting, and the three-round syndrome-history decoder. Temporal decoding improves over raw majority voting, but the additional syndrome-extraction rounds introduce substantial circuit overhead.

Key observation. This experiment reveals an important distinction between *decoder quality* and *fault-tolerant circuit efficiency*. The syndrome-history decoder was demonstrably better than raw majority voting because it exploited time-resolved information about changing errors. Nevertheless, obtaining that information required three complete noisy stabilizer-extraction rounds. In the present distance-three implementation, the additional fault opportunities introduced by those rounds outweighed the decoding improvement. Therefore,

$$\boxed{\text{more syndrome information does not necessarily imply lower logical error}} \quad (39)$$

when the process used to acquire that information is itself noisy. This result motivated the use of the lower-overhead adaptive flagged decoder in the final logical-gate and end-to-end FT models.

10 Gate-Aware End-to-End Fault-Tolerant Model

Motivation. The preceding experiments focused primarily on the error-correction subsystem. However, a complete fault-tolerant quantum computation must also account for the cost of performing logical operations on encoded qubits. Treating every logical gate as if it were followed by the same single error-correction channel can substantially underestimate the accumulated error of a realistic logical circuit. This issue is particularly important for the present variational quantum classifier because each inference contains repeated arbitrary rotations and entangling operations. We therefore introduced a gate-aware logical model in which the previously characterized protected logical channel is composed multiple times to represent the additional fault-tolerant overhead associated with logical gate gadgets.

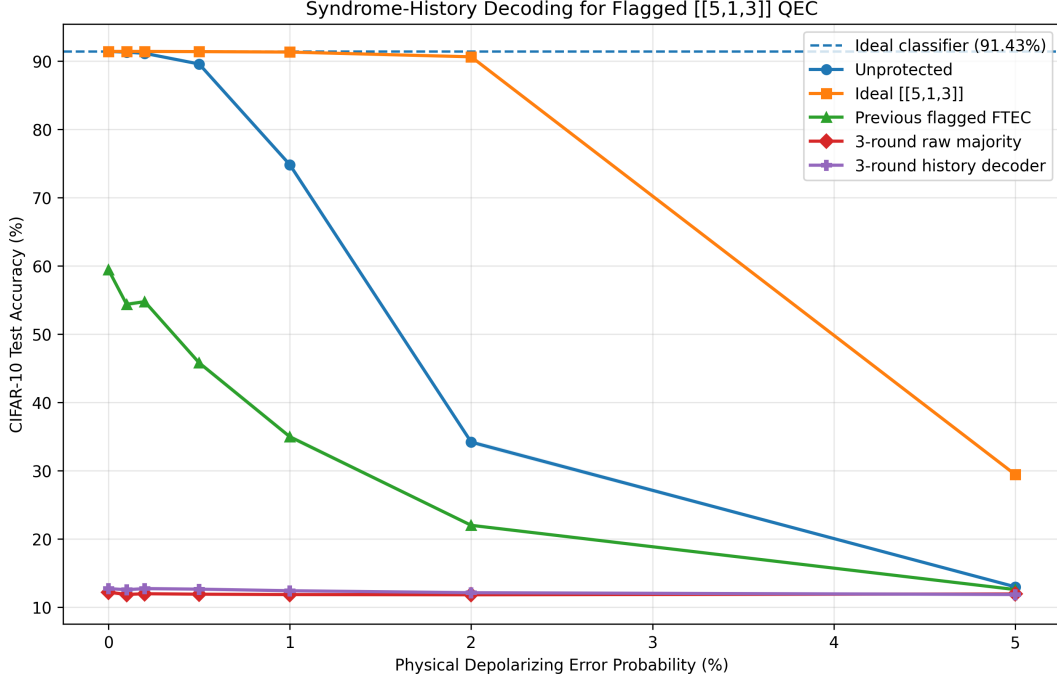


Figure 17: CIFAR-10 test accuracy for the unprotected classifier, ideal $[[5, 1, 3]]$ protection, previous adaptive flagged FTEC, three-round raw majority voting, and three-round syndrome-history decoding. The history decoder improves slightly over raw majority voting but remains substantially below the adaptive flagged strategy because three complete syndrome rounds add significant circuit noise.

Fixed decoder selection. Before constructing the final model, a single error-correction strategy was selected and held fixed over the entire physical-noise sweep. The three candidate decoders were the adaptive flagged-FTEC procedure, the three-round syndrome-history decoder, and the three-round raw-majority decoder. Their mean logical error probabilities over the complete tested range were

$$\overline{p}_L^{\text{flagged}} = 0.023167, \quad \overline{p}_L^{\text{history}} = 0.062830, \quad \overline{p}_L^{\text{majority}} = 0.083861. \quad (40)$$

The adaptive flagged-FTEC decoder therefore provided the smallest average logical error and was selected as the fixed decoder for all subsequent logical-gate and end-to-end experiments. Importantly, the decoder was selected globally rather than independently at each value of p , avoiding an unrealistically favorable per-noise-point choice.

Selected protected logical channel. The logical error probabilities produced by the selected flagged decoder were

$$p_L = [0.013274, 0.014274, 0.014294, 0.016382, 0.019848, 0.027324, 0.056772] \quad (41)$$

for physical depolarizing probabilities

$$p = [0, 0.001, 0.002, 0.005, 0.01, 0.02, 0.05], \quad (42)$$

respectively. Because the syndrome-history result files retained only the total residual logical error probability, the final logical channel was represented symmetrically as

$$\mathcal{E}_L(\rho) = (1 - p_L)\rho + \frac{p_L}{3} (X_L \rho X_L + Y_L \rho Y_L + Z_L \rho Z_L). \quad (43)$$

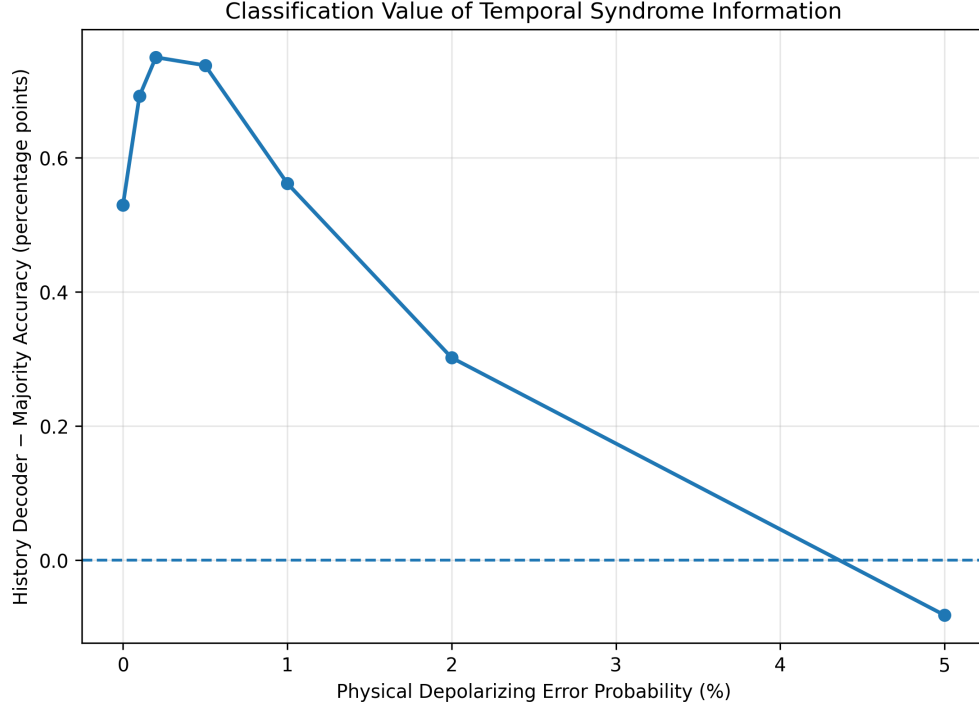


Figure 18: Difference in CIFAR-10 accuracy between the three-round syndrome-history decoder and raw three-round majority voting. Temporal information provides a modest benefit over majority voting over most of the tested noise range, although this benefit disappears at the highest physical error rate.

This approximation preserves the measured total logical-error probability while avoiding unsupported assumptions about a particular residual logical Pauli bias.

Logical circuit structure. The trained classifier contains four data-reuploading layers over eight logical qubits. Each layer contains eight data-encoding R_Y rotations, eight trainable R_Y rotations, eight trainable R_Z rotations, and an eight-gate nearest-neighbor CNOT ring. Consequently, one inference contains

$$N_{R_Y}^{\text{data}} = 32, \quad N_{R_Y}^{\text{train}} = 32, \quad N_{R_Z}^{\text{train}} = 32, \quad N_{\text{CNOT}} = 32. \quad (44)$$

Thus, the classifier uses

$$N_{\text{rot}} = 96 \quad (45)$$

arbitrary single-qubit rotations and 32 logical entangling gates. Counting the two logical-qubit participants of every CNOT gives

$$N_{\text{CNOT,part}} = 64, \quad (46)$$

and the uniform logical-noise model therefore contains

$$96 + 64 = 160 \quad (47)$$

logical fault locations per inference.

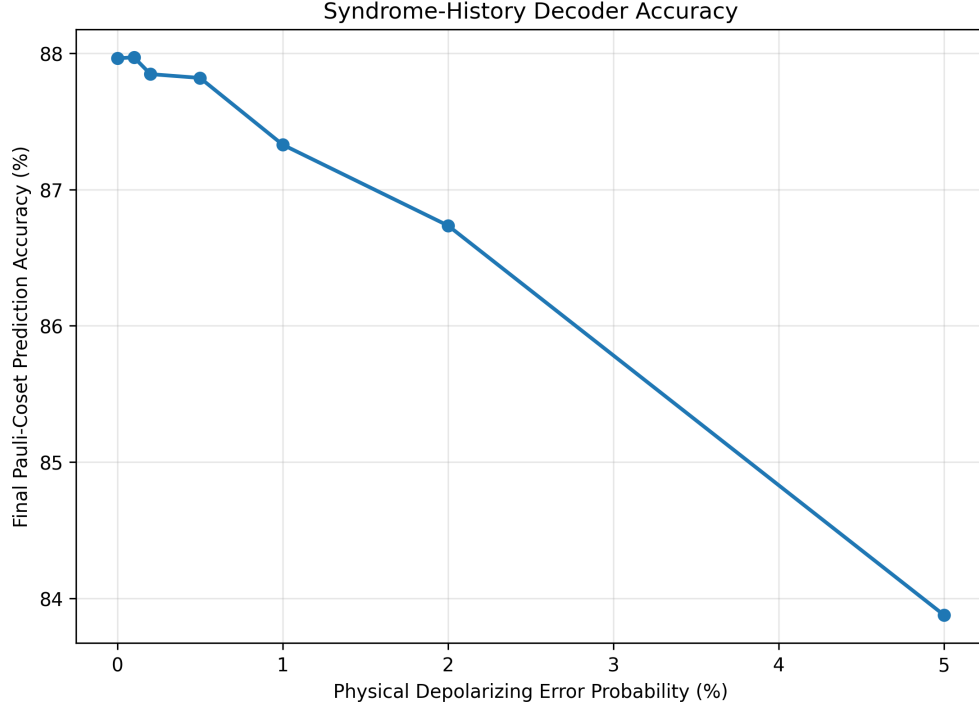


Figure 19: Accuracy of the syndrome-history maximum-likelihood decoder in predicting the final $[[5, 1, 3]]$ Pauli coset. Prediction accuracy remains high but decreases gradually as the physical depolarizing error probability increases.

Why gate-aware overhead is necessary. The arbitrary R_Y and R_Z gates used by the classifier cannot simply be assumed to have the same implementation cost as one protected logical Pauli location. Similarly, an encoded logical entangling gate requires more structure than an ideal bare CNOT. Rather than claiming an exact physical compilation at this stage, we introduced a resource-aware logical-gadget model. One application of the selected flagged-FTEC logical channel was treated as one protected error-correction-equivalent cycle. The final model assigned

$$n_{\text{rot}} = 2 \quad (48)$$

such cycles to every arbitrary logical rotation and

$$n_{\text{CNOT}} = 2 \quad (49)$$

cycles to each logical-qubit participant of a CNOT. The resulting number of protected channel applications per inference was therefore

$$N_{\text{FT}} = 96 \times 2 + 64 \times 2 = 320. \quad (50)$$

These values are used as a controlled overhead proxy rather than as exact hardware gate counts.

Channel composition. If a protected logical Pauli channel with probability vector

$$\mathbf{q} = [P(I_L), P(X_L), P(Y_L), P(Z_L)] \quad (51)$$

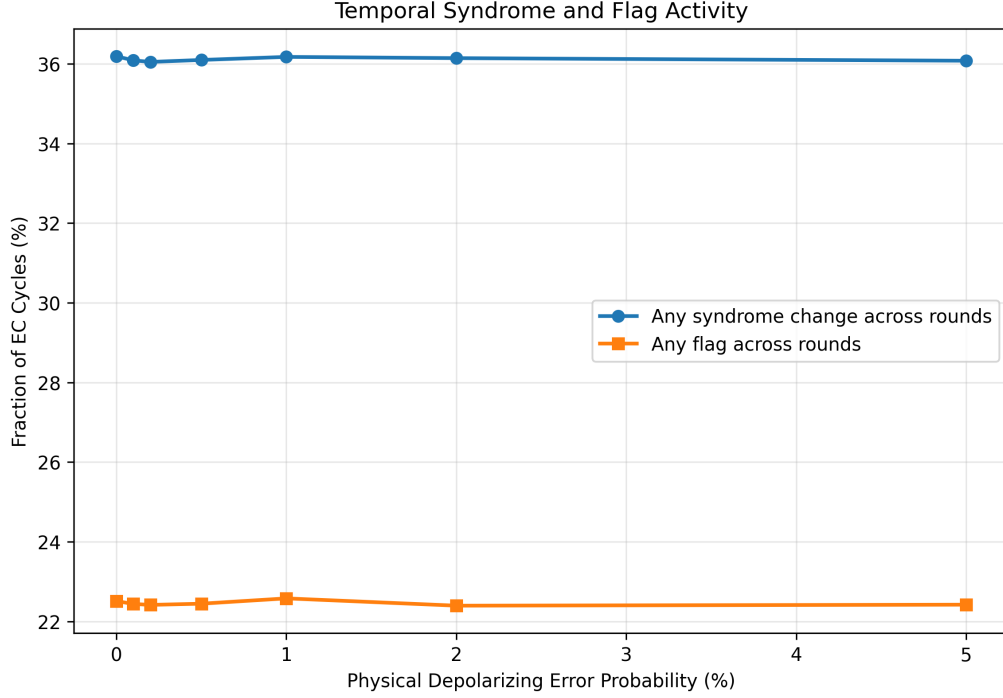


Figure 20: Fraction of three-round error-correction cycles containing at least one syndrome change or at least one flag event. Temporal syndrome changes occur frequently because the noisy extraction circuitry can modify the data state between rounds.

is incurred repeatedly, the effective error channel of a logical gadget is obtained by composing the Pauli channel with itself. For the two-cycle model,

$$\mathcal{E}_{\text{gadget}} = \mathcal{E}_L \circ \mathcal{E}_L. \quad (52)$$

At $p = 0$, for example, the selected correction procedure has

$$p_L = 0.013274, \quad (53)$$

while the corresponding two-cycle rotation and CNOT-participant gadgets have

$$p_L^{\text{gadget}} = 0.026313. \quad (54)$$

At $p = 0.01$, these values become

$$p_L = 0.019848, \quad p_L^{\text{gadget}} = 0.039171. \quad (55)$$

The complete set of gate-aware logical error probabilities is shown in Fig. 21.

Accumulated logical-fault burden. Although the single-gadget logical-error probabilities remain only a few percent in the low-noise regime, the classifier contains many such locations. A useful summary statistic is the expected number of non-identity logical Pauli events per inference,

$$\mathbb{E}[N_{\text{fault}}] = N_{\text{rot}} p_L^{\text{rot}} + N_{\text{CNOT,part}} p_L^{\text{CNOT}}. \quad (56)$$

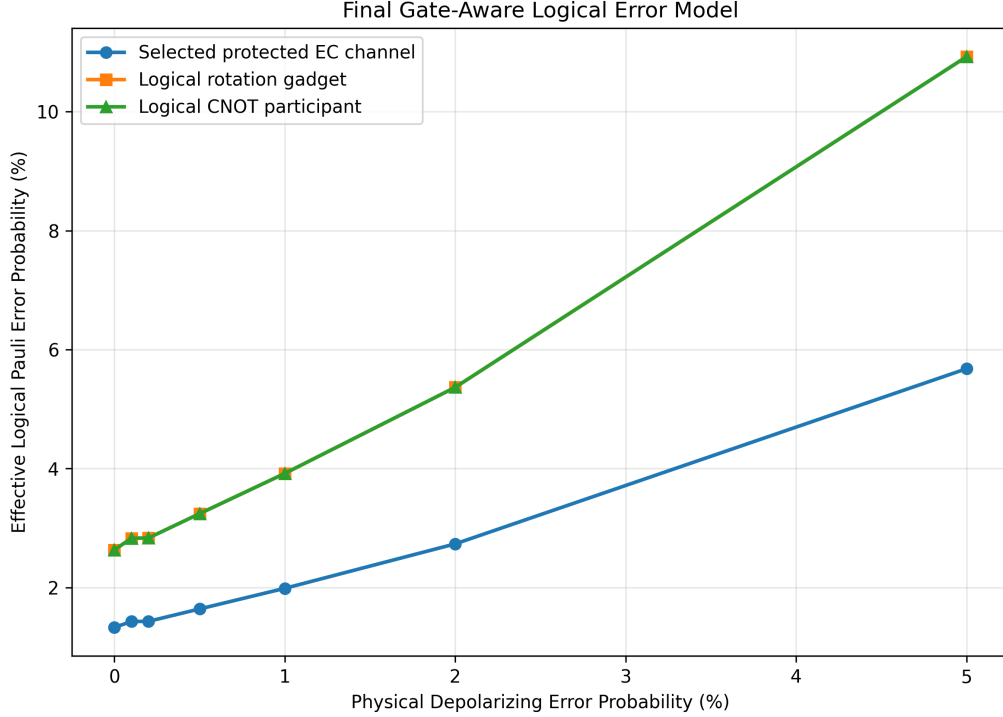


Figure 21: Effective logical error probabilities used in the final gate-aware model. The selected protected error-correction channel is composed twice to represent the additional overhead of arbitrary logical rotations and logical CNOT participants.

The resulting values are reported in Table 8. Even when the externally swept physical data-noise probability is $p = 0$, the noisy fault-tolerant circuitry produces an expected

$$\mathbb{E}[N_{\text{fault}}] = 4.210 \quad (57)$$

non-identity logical events during one classifier inference. This increases to 6.267 events at $p = 0.01$ and 17.479 events at $p = 0.05$.

Intermediate gate-aware experiment. Before freezing the final model, we evaluated the consequence of applying the three-round syndrome-history channel to the logical-gate overhead model. This experiment produced extremely poor classification performance because the history decoder itself had a large logical-error floor,

$$p_L^{\text{history}}(p = 0) = 0.053720. \quad (58)$$

After two-cycle gate composition, the effective logical gadget error increased to

$$p_L^{\text{gadget}} \approx 0.103592. \quad (59)$$

Consequently, the uniform-history classifier achieved only 12.63% accuracy at $p = 0$, while the corresponding gate-aware model achieved 11.68%. Figure 23 shows that this configuration remained near the random-classification regime throughout the tested noise range. This result motivated the global decoder-selection procedure described above and confirmed that a sophisticated decoder cannot compensate for excessive error-correction overhead if its own logical-error floor is too large.

Table 8: Final gate-aware logical error rates and expected accumulated logical-fault burden.

p	Base p_L	Gadget p_L	$\mathbb{E}[N_{\text{fault}}]$
0.000	0.013274	0.026313	4.210
0.001	0.014274	0.028276	4.524
0.002	0.014294	0.028316	4.530
0.005	0.016382	0.032406	5.185
0.010	0.019848	0.039171	6.267
0.020	0.027324	0.053653	8.584
0.050	0.056772	0.109247	17.479

Table 9: Final end-to-end CIFAR-10 test accuracy. Results are mean $\pm$ standard deviation over five stochastic seeds.

p	Unprotected	UniformProtected	FinalFT
0.000	0.9143 $\pm$ 0.0000	0.5890 $\pm$ 0.0039	0.2352 $\pm$ 0.0017
0.001	0.9132 $\pm$ 0.0004	0.5434 $\pm$ 0.0019	0.2108 $\pm$ 0.0025
0.002	0.9115 $\pm$ 0.0007	0.5410 $\pm$ 0.0036	0.2087 $\pm$ 0.0027
0.005	0.8959 $\pm$ 0.0024	0.4535 $\pm$ 0.0036	0.1802 $\pm$ 0.0027
0.010	0.7483 $\pm$ 0.0019	0.3478 $\pm$ 0.0047	0.1506 $\pm$ 0.0031
0.020	0.3419 $\pm$ 0.0038	0.2190 $\pm$ 0.0035	0.1259 $\pm$ 0.0031
0.050	0.1299 $\pm$ 0.0017	0.1240 $\pm$ 0.0020	0.1181 $\pm$ 0.0032

Sensitivity to rotation-gadget overhead. The intermediate logical-gate experiment also varied the number of protected cycles assigned to arbitrary rotations from one to three. Figure 24 shows that the resulting accuracy differences were small once the classifier had already entered the noise-saturated regime. This should not be interpreted as evidence that arbitrary logical rotations are inexpensive; rather, the underlying syndrome-history channel was already sufficiently destructive that additional rotation overhead caused little further measurable degradation.

Final end-to-end comparison. The final experiment combined the globally selected adaptive flagged decoder with the gate-aware overhead model. Three configurations were evaluated: an unprotected noisy classifier, a *UniformProtected* model that applied the selected flagged logical channel once at each of the 160 modeled logical locations, and the *FinalFT* model that used the two-cycle logical-gadget overhead described above. The results are reported in Table 9 and Fig. 25.

Effect of the protected error floor. A striking result is that even at $p = 0$, the UniformProtected classifier achieved only

$$58.90\% \tag{60}$$

accuracy, while the gate-aware FinalFT model achieved

$$23.52\%. \tag{61}$$

This behavior follows directly from the nonzero error-correction floor. In this experiment, $p = 0$ removes only the externally swept physical data-depolarizing error; the syndrome extraction, ancilla, measurement, recovery, and logical-gadget noise parameters remain nonzero. Therefore, the protected models continue to accumulate logical faults even at the leftmost point of the sweep.

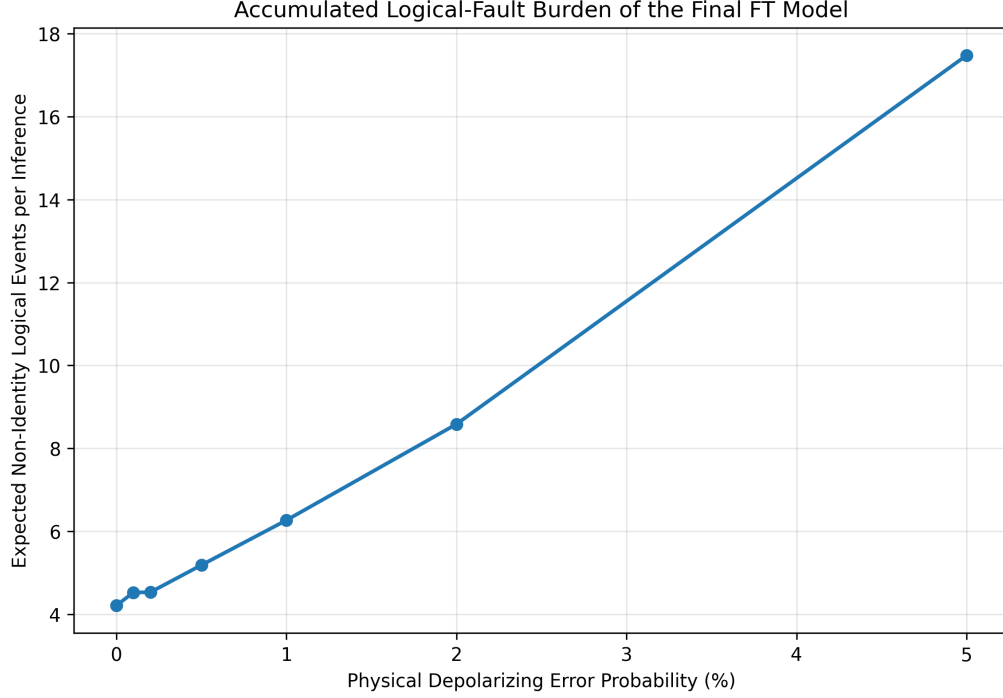


Figure 22: Expected number of non-identity logical Pauli events accumulated during one classifier inference under the final gate-aware fault-tolerant model. Even at zero externally applied data noise, noisy QEC and logical-gate overhead produce several expected logical fault events.

Fault-tolerant overhead penalty. Figure 26 isolates the additional loss caused by moving from one protected logical channel per location to the gate-aware logical-gadget model,

$$\Delta A_{\text{overhead}} = A_{\text{FinalFT}} - A_{\text{UniformProtected}}. \quad (62)$$

The penalty is largest in the low-noise regime, where UniformProtected still retains meaningful classification ability. As the physical noise increases, both protected configurations eventually approach the low-accuracy regime and the absolute difference narrows.

No end-task break-even point in the tested range. To determine whether the fault-tolerant architecture ever provided an end-task advantage, we evaluated

$$\Delta A_{\text{FT}} = A_{\text{FinalFT}} - A_{\text{unprotected}}. \quad (63)$$

As shown in Fig. 27, ΔA_{FT} remained negative over the entire tested physical-noise range. Consequently, no FinalFT/unprotected accuracy break-even point was observed for

$$0 \leq p \leq 0.05. \quad (64)$$

At low physical error, the unprotected circuit strongly outperformed the FT model because the correction overhead dominated. At high physical error, both models approached the low-accuracy regime before an FT advantage could emerge.

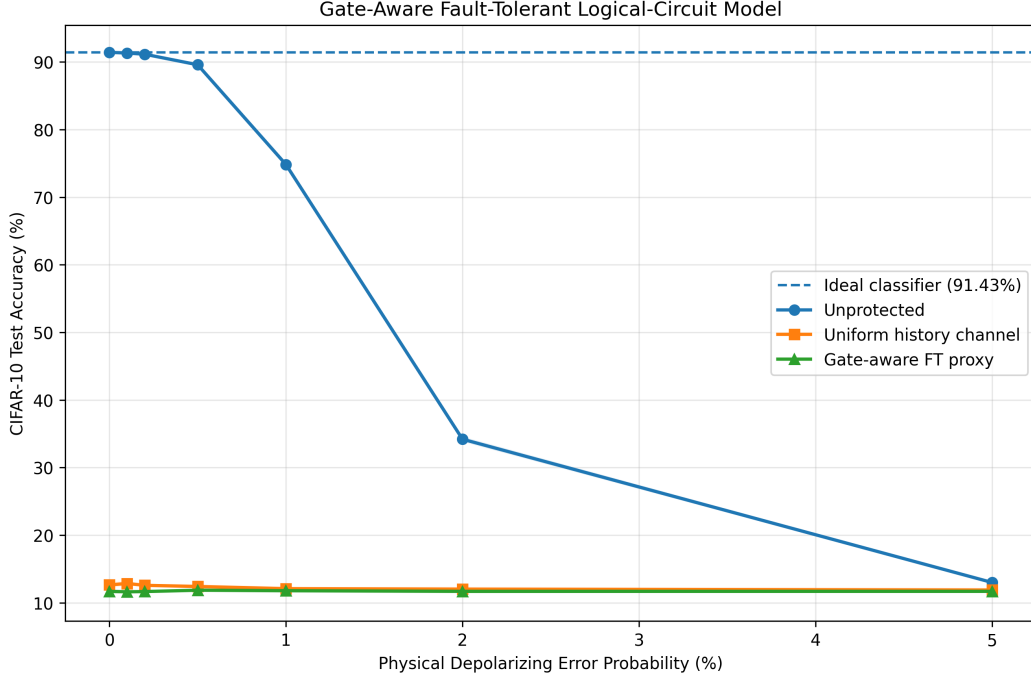


Figure 23: Gate-aware logical-gate experiment using the three-round syndrome-history channel. Because the history channel already has a substantial logical-error floor, both the uniform and gate-aware protected classifiers operate close to the low-accuracy regime.

77% target-accuracy analysis. A target CIFAR-10 accuracy of

$$A_{\text{target}} = 77\% \quad (65)$$

was used as an application-level operating requirement. The unprotected classifier remained above this threshold through the tested physical error probability

$$p = 0.005, \quad (66)$$

where its mean accuracy was 89.59%. At $p = 0.01$, the unprotected accuracy decreased to 74.83% and therefore fell below the target. In contrast, the FinalFT model did not satisfy the 77% requirement at any tested physical error probability, including $p = 0$, because its gate-aware correction overhead already reduced the accuracy to 23.52%.

Resource estimate. The final model used eight logical qubits protected by the $[[5, 1, 3]]$ code, corresponding to

$$8 \times 5 = 40 \quad (67)$$

physical data qubits. Assuming one syndrome ancilla and one flag ancilla can be allocated to each logical block simultaneously gives an additional

$$8 \times 2 = 16 \quad (68)$$

ancillas and the conceptual blockwise estimate

$$N_{\text{qubits}} \approx 56. \quad (69)$$

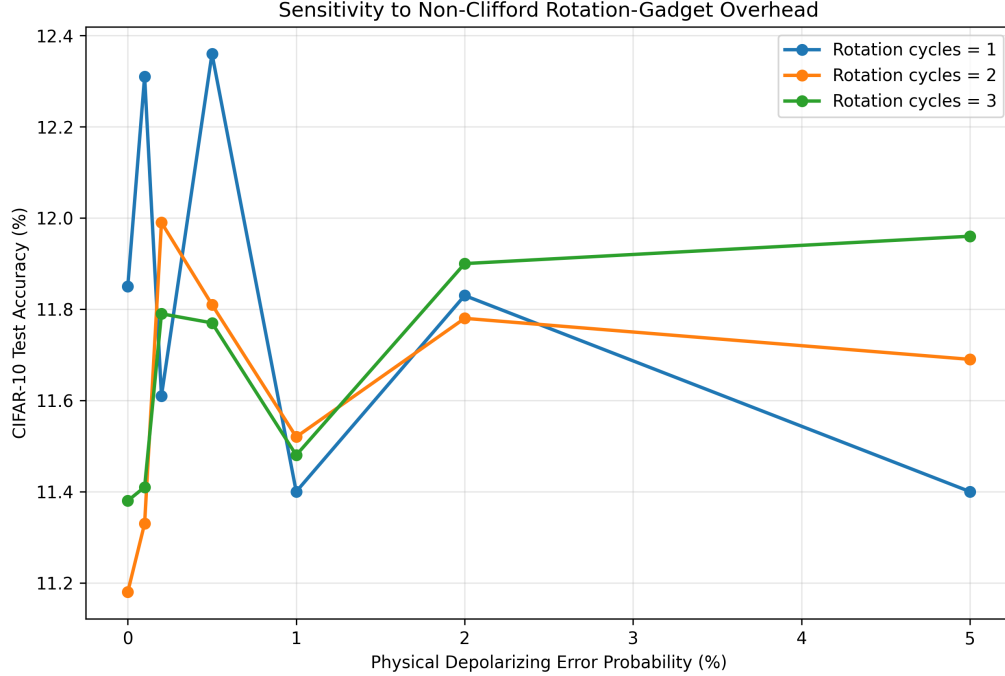


Figure 24: Sensitivity of CIFAR-10 accuracy to the number of protected cycles assigned to arbitrary logical rotations in the intermediate syndrome-history model. Once the classifier is already strongly noise-saturated, further increases in rotation overhead produce only small additional accuracy changes.

This value should not be interpreted as an exact hardware requirement. A compiled physical implementation may reuse ancillas sequentially, require additional ancillas for specific logical-gate gadgets, or incur connectivity-dependent routing overhead. The value of 56 is therefore used only to communicate the approximate scale of the encoded architecture.

Interpretation. The final experiment reveals that the principal challenge is no longer whether the $[[5, 1, 3]]$ code can suppress isolated physical errors. Under ideal QEC, it clearly can. The limiting factor is the accumulated cost of performing error correction and logical gates often enough to protect a comparatively deep variational circuit. In the present model, the adaptive flagged decoder substantially reduces hook-error propagation, but its remaining logical error floor is still too large to be applied hundreds of times during one classifier inference. Thus, the final result is

$$\boxed{\text{fault-tolerant protection is beneficial only if the logical error per protected operation is sufficiently below the inv}} \quad (70)$$

for the target application.

Main end-to-end finding. The complete progression from ideal QEC to gate-aware fault tolerance therefore changes the interpretation of the classifier dramatically. Ideal $[[5, 1, 3]]$ protection preserves more than 90% CIFAR-10 accuracy at physical noise levels for which the unprotected model has already degraded severely. However, once noisy syndrome extraction, correlated fault propagation, decoder overhead, and logical-gate overhead are incorporated, the same small-code FT architecture no longer provides an application-level advantage. The result does not indicate a failure of QEC in

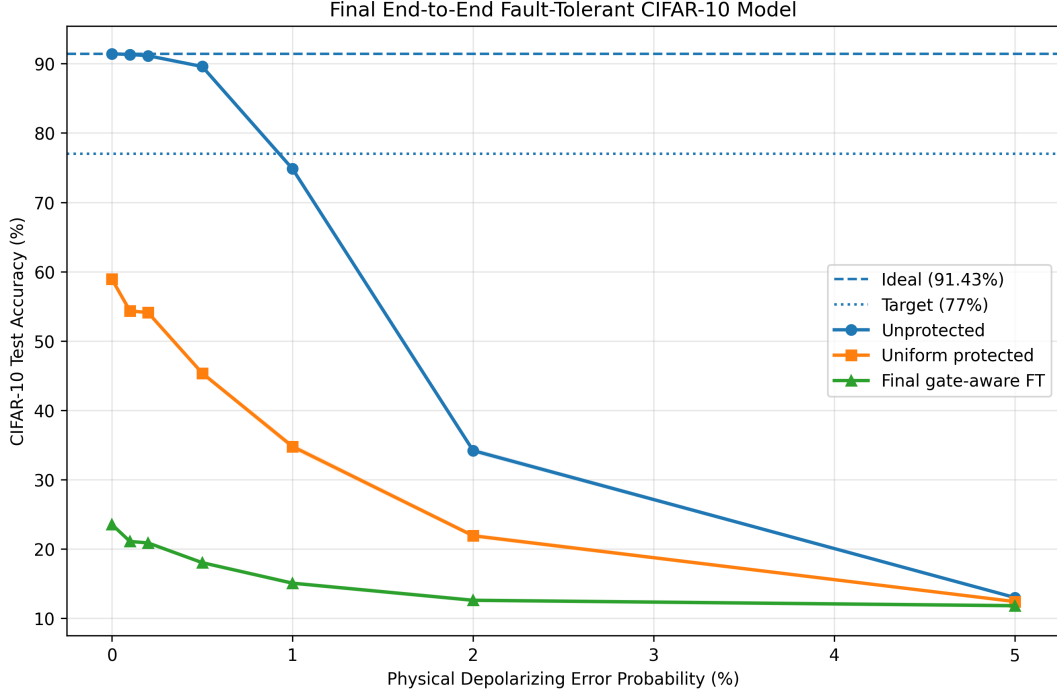


Figure 25: Final end-to-end CIFAR-10 accuracy for the unprotected classifier, the uniform protected logical-channel model, and the gate-aware fault-tolerant model. The ideal 91.43% classifier baseline and the 77% target accuracy are shown for reference.

principle; rather, it quantifies the stringent logical-error and overhead requirements that must be met before fault-tolerant protection becomes advantageous for a nontrivial quantum machine-learning circuit.

11 Discussion and Lessons Learned

Overview. The experiments in this study were intentionally constructed as a progression from idealized quantum error correction to increasingly realistic fault-tolerant operation. This progression was necessary because the behavior observed under ideal QEC was qualitatively different from the behavior of the same code once syndrome extraction, recovery, repeated measurements, and logical-gate overhead were allowed to fail. The main lesson is therefore not simply whether QEC improved CIFAR-10 classification, but *which assumptions were responsible for the apparent improvement or degradation*. The experiments demonstrate that code distance alone is insufficient for predicting application-level performance; fault propagation, syndrome-circuit design, decoder strategy, circuit depth, and logical-gate overhead must be considered together.

Ideal QEC provides an optimistic upper bound. Under ideal $[[5, 1, 3]]$ protection, the classifier exhibited strong robustness to general Pauli noise. At $p = 0.01$, for example, the unprotected accuracy decreased to 74.74%, while ideal QEC retained 91.30%. At $p = 0.02$, the contrast became even larger, with the unprotected classifier reaching only 34.43% while the ideal protected model maintained 90.64%. These results show that the five-qubit code is in principle capable of converting the physical error rates considered in this work into logical error probabilities small enough to

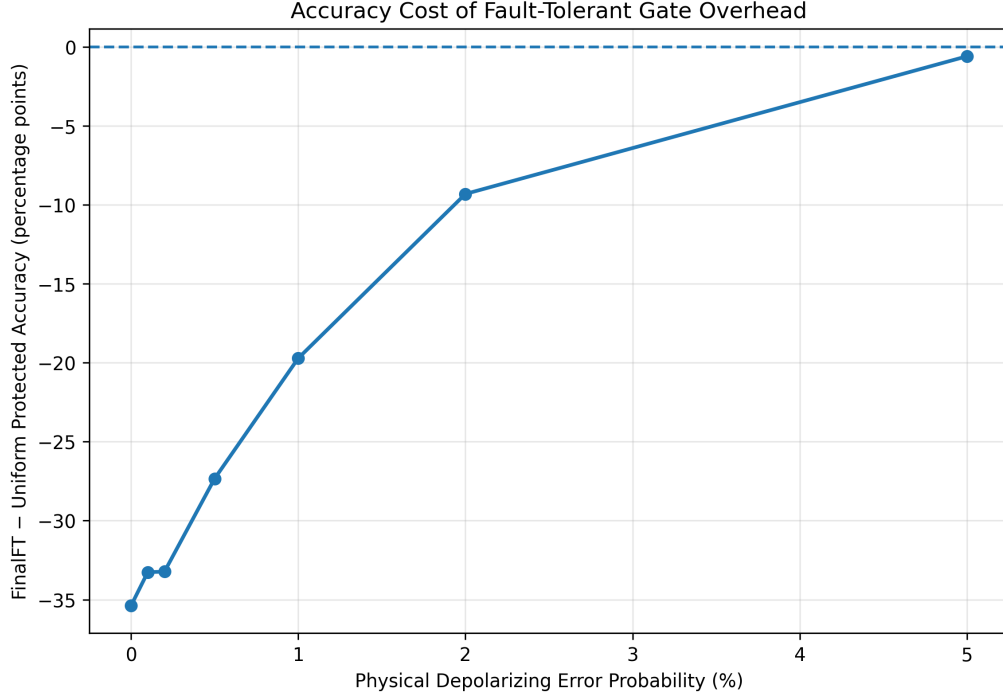


Figure 26: Additional CIFAR-10 accuracy penalty caused by gate-aware fault-tolerant logical-gadget overhead relative to the simpler uniform protected-channel model.

preserve the trained model. However, the ideal experiment excludes the cost of actually obtaining the syndrome and applying a recovery. It should therefore be interpreted as an upper bound rather than a realistic FTQC prediction.

QEC and fault tolerance are fundamentally different requirements. The largest conceptual transition occurred when the error-correction circuit itself became noisy. Under naive single-ancilla stabilizer extraction, the logical error probability at zero externally applied data noise increased to

$$p_L^{\text{naive}}(0) = 0.039094. \quad (71)$$

This result is particularly informative because the physical data channel itself contributes no swept error at this point. The logical error arises entirely from the correction circuitry. The resulting CIFAR-10 accuracy of approximately 15% shows that applying a theoretically valid error-correcting code through a non-fault-tolerant circuit can be substantially worse than leaving the logical circuit unprotected.

Fault propagation is more important than the presence of noise alone. The earlier repetition-code transversal-CNOT experiment and the later five-qubit flagged experiment both demonstrated the importance of limiting how far one physical fault can spread. In the repetition-code experiment, transversal coupling changed the observed gate-error scaling from approximately first order for the non-transversal construction to close to second order for the transversal construction. In the five-qubit experiment, a flag ancilla reduced the logical-error floor from

$$0.039094 \quad (72)$$

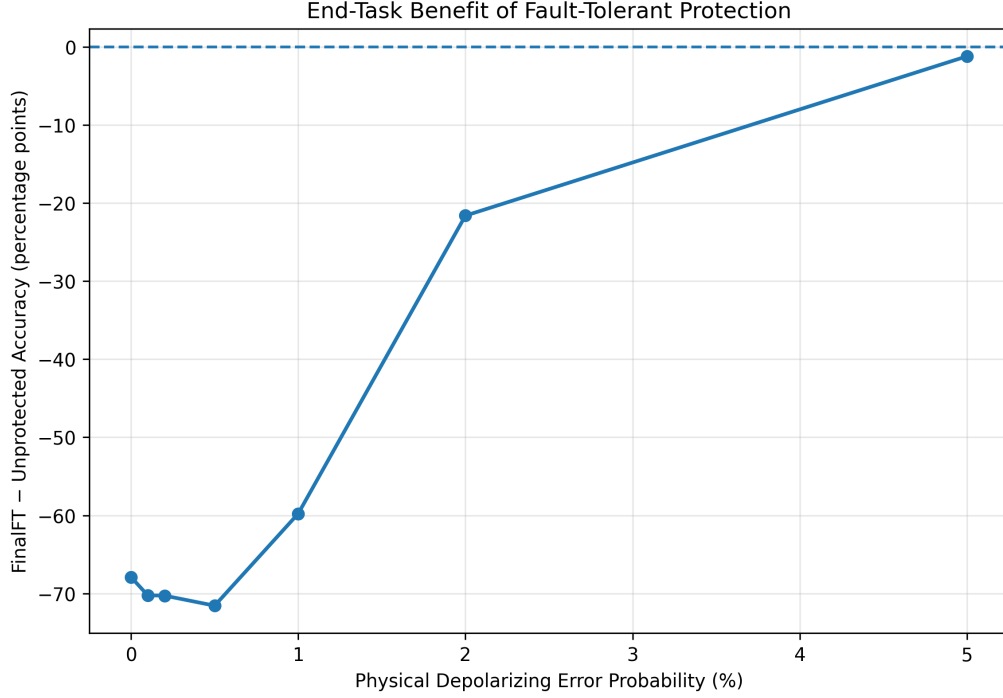


Figure 27: FinalFT accuracy relative to the unprotected classifier. No positive end-task break-even point is observed over the tested physical-noise range.

to

$$0.013274, \quad (73)$$

a reduction of approximately 66%. The physical gate-error probabilities were not reduced; instead, the circuit was modified so that dangerous correlated faults became detectable. This distinction is central to fault-tolerant circuit design.

Repeated syndrome extraction exhibits a noise-information tradeoff. Repeating syndrome measurements initially appeared beneficial in the repetition-code experiments, particularly when syndrome-bit noise was dominant and the underlying data error was fixed. In that setting, three measurements followed by majority voting substantially improved syndrome reliability. Once data faults could occur during the repeated extraction process, however, the situation changed. A sequence of syndromes then contains information about both measurement faults and newly created data faults. The syndrome-history decoder exploited this temporal structure and reduced the mean logical error relative to raw majority voting,

$$0.083861 \rightarrow 0.062830. \quad (74)$$

Nevertheless, both three-round strategies remained inferior to the lower-overhead adaptive flagged decoder, whose mean logical error was only 0.023167. Thus, repeated extraction illustrates a general FTQC tradeoff:

$$\boxed{\text{additional syndrome information} \quad \text{versus} \quad \text{additional fault opportunities.}} \quad (75)$$

Adaptive correction was preferable to unconditional repetition. The branch statistics of the flagged decoder help explain why it performed better. At $p = 0$, approximately 82.16% of correction cycles required no expensive follow-up path. Even at $p = 0.01$, the no-trigger branch remained active in approximately 78.20% of cases. The adaptive strategy therefore acquired additional information only when a flag or nontrivial syndrome indicated that it was needed. In contrast, the syndrome-history decoder paid the cost of three complete rounds for every block. For the present small distance-three code and circuit-noise parameters, adaptive acquisition of syndrome information was therefore substantially more efficient than unconditional repetition.

Circuit depth converts small logical errors into large application-level degradation. A major reason for the large difference between block-level and classifier-level results is the number of logical fault opportunities. The VQC contains 96 arbitrary rotations and 32 logical CNOTs. In the uniform logical model, this corresponds to 160 logical-qubit fault locations. The final gate-aware approximation increased the corresponding protected-channel count to 320 EC-equivalent cycles. Even a per-location logical error probability of approximately 1% is therefore not small at the algorithm level. For example, the selected flagged decoder has

$$p_L(0) = 0.013274, \quad (76)$$

which may appear modest in isolation. However, after gate-aware composition and repeated use across the full classifier, the model predicts approximately

$$4.21 \quad (77)$$

non-identity logical Pauli events per inference even at zero externally swept data noise. Application-level robustness therefore requires a logical error probability considerably smaller than what appears acceptable when examining a single encoded block.

The logical error rate must be considered relative to circuit scale. The preceding observation suggests an approximate design principle. If a circuit contains N protected logical fault opportunities, then maintaining a high probability of completing the circuit without a logical fault requires the effective logical error probability to satisfy roughly

$$Np_L \ll 1. \quad (78)$$

This condition is not a strict performance theorem for the present classifier because different Pauli faults have different effects and some faults can partially cancel. Nevertheless, it provides a useful interpretation of the numerical results. For N on the order of several hundred, a logical error probability near 10^{-2} remains too large for reliable end-to-end inference.

Application-level evaluation changes the interpretation of QEC performance. If this study had stopped at the encoded-block level, the reduction

$$p = 0.01 \quad \rightarrow \quad p_L^{\text{ideal}} \approx 9.78 \times 10^{-4} \quad (79)$$

would suggest excellent protection. If it had stopped after introducing flags, the reduction of the noisy-QEC logical error by approximately 66% would likewise appear highly encouraging. The CIFAR-10 experiments show why an application-level metric is necessary: improvements that are substantial at the encoded-block level may still be insufficient when the protected operation is repeated many times. Classification accuracy therefore serves as an end-to-end measure of whether physical error suppression is actually strong enough for the target workload.

The absence of a final break-even point is informative. No physical-noise value in the tested range produced

$$A_{\text{FinalFT}} > A_{\text{unprotected}}. \quad (80)$$

Similarly, the FinalFT model never reached the target accuracy of 77%. This negative result is meaningful because simpler versions of the experiment did show strong QEC gains. The disappearance of that benefit as increasingly realistic overhead was introduced identifies the limiting component of the current architecture. The issue is not that the $[[5, 1, 3]]$ code cannot correct isolated physical errors; rather, the combination of noisy repeated correction and the number of logical operations causes the total logical-fault burden to exceed the tolerance of the classifier.

Implications for quantum machine-learning circuit design. The results suggest that fault-tolerant quantum machine learning cannot be treated as a post-processing step in which an arbitrary variational circuit is designed first and QEC is added afterward. The logical circuit itself should be designed jointly with the error-correcting architecture. Shallower circuits, fewer non-Clifford rotations, fewer repeated data-uploading layers, and logical operations naturally compatible with the selected code could substantially reduce the number of expensive protected operations. Thus, a fault-tolerance-aware variational ansatz may be more appropriate than directly protecting a NISQ-style ansatz.

Overall lesson. The complete experimental sequence supports the following interpretation:

ideal QEC can strongly suppress physical errors,
 fault-tolerant extraction is necessary to preserve that benefit,
 decoder information must justify its acquisition cost, and
 logical-gate overhead must remain small relative to circuit depth.

(81)

The central challenge is therefore not merely to lower the physical error probability, but to minimize the product of logical error per protected operation and the number of protected operations required by the application.

12 Limitations and Hardware Translation

Logical-level rather than full encoded-state simulation. The largest computational approximation in this work is the separation between the eight-qubit logical statevector and the physical error-correction model. Eight logical qubits encoded by the $[[5, 1, 3]]$ code correspond to at least 40 physical data qubits, before ancillas are included. An explicit pure-state simulation would require a Hilbert space of size

$$2^{40}, \quad (82)$$

and would become still larger when syndrome and flag ancillas were included. Instead, physical faults and stabilizer information were propagated using Pauli frames, while the machine-learning circuit was evaluated on an eight-qubit logical statevector. This approach enables large-scale CIFAR-10 evaluation but does not capture arbitrary coherent correlations across the complete physical encoded state.

Pauli noise approximation. The physical noise models considered in the experiments are stochastic Pauli or depolarizing channels. These models are useful for studying QEC because stabilizer codes admit efficient Pauli-frame simulation, but real quantum processors may exhibit coherent over-rotations, amplitude damping, dephasing asymmetry, leakage, crosstalk, time-dependent drift, and spatially correlated noise. Such effects can produce logical behavior that differs from a symmetric depolarizing approximation. Hardware validation is therefore required before interpreting the simulator results as predictions of a specific physical processor.

Fixed circuit-noise parameters. The noisy-QEC experiments used

$$p_{2Q} = 0.005, \quad p_{1Q} = 0.001, \quad p_{\text{prep}} = 0.001, \quad p_{\text{meas}} = 0.01, \quad p_{\text{rec}} = 0.005. \quad (83)$$

These values were selected to create a controlled and interpretable circuit-level fault model. They are not calibrated to a particular Rigetti processor. Consequently, quantities such as the 1.3274% flagged logical-error floor should be interpreted as results of this simulation configuration rather than hardware-specific logical error rates.

Simplified residual logical channel. The earlier five-qubit simulations retained the full residual distribution

$$P(I_L), \quad P(X_L), \quad P(Y_L), \quad P(Z_L). \quad (84)$$

However, the final syndrome-history result file retained only the total logical error probability. The final gate-aware model therefore used the symmetric approximation

$$P(X_L) = P(Y_L) = P(Z_L) = \frac{p_L}{3}. \quad (85)$$

If the actual decoder produces a strongly biased logical channel, this approximation can alter the predicted classifier response because the trained VQC may not be equally sensitive to X_L , Y_L , and Z_L faults. Future simulations should retain the complete logical Pauli distribution through every decoding stage.

Approximate logical-rotation model. The arbitrary R_Y and R_Z operations in the trained VQC were not explicitly compiled into an exact universal fault-tolerant gate sequence. Instead, each logical rotation was assigned two protected EC-equivalent cycles. This provides a controlled way to quantify the effect of logical-gate overhead but does not model the complete resources of rotation synthesis, state injection, magic-state preparation, or another universal logical-gate construction. Therefore, the FinalFT accuracy should be interpreted as a gate-aware logical proxy rather than an exact implementation of a universal fault-tolerant processor.

Approximate logical entangling-gate model. Similarly, logical CNOT operations were modeled by applying the protected logical channel independently to the control and target participants with a two-cycle overhead. A complete pieceable fault-tolerant implementation can create correlated two-block errors and can require intermediate error correction on a joint encoded system. Such correlations are not represented by independent single-logical-qubit Pauli channels. Explicit simulation of the entangling gadget is therefore an important extension of the current model.

Recovery model. The simulations retained a nonzero recovery fault probability,

$$p_{\text{rec}} = 0.005, \quad (86)$$

when a physical recovery operation was executed. In many practical QEC architectures, corrections can instead be tracked in a classical Pauli frame and propagated forward without physically applying the Pauli gate immediately. A Pauli-frame implementation could therefore remove part of the modeled recovery overhead. Evaluating both physical and frame-tracked correction strategies would provide a useful future ablation.

Finite stochastic trajectories. The CIFAR-10 noise experiments averaged

$$32 \quad (87)$$

Monte Carlo trajectories per input and reported results over five random seeds. This provided stable trends while keeping the computational cost practical on an RTX 4090 GPU. Nevertheless, some residual Monte Carlo variation remains. Increasing the trajectory count would reduce stochastic uncertainty, particularly near break-even regions where relatively small accuracy differences are important.

No retraining under noise. The quantum-classical classifier was trained under ideal quantum simulation and subsequently evaluated under noise. The variational parameters and classical output head were not fine-tuned using the protected logical channel. Noise-aware training or logical-noise-aware fine-tuning could potentially improve the robustness of the classifier. However, such retraining would answer a different research question: the present experiments intentionally measure how well QEC preserves an already trained ideal model.

No claim of quantum advantage. The CIFAR-10 system is a hybrid quantum-classical model. A pretrained ResNet18 extracts a learned 32-dimensional representation, while the eight-qubit variational circuit and classical head perform the subsequent classification. The 91.43% baseline therefore demonstrates that the hybrid pipeline is sufficiently accurate to study end-task fault tolerance; it does not establish a computational or statistical quantum advantage over classical image classifiers.

Conceptual qubit count. Eight logical qubits require

$$40 \quad (88)$$

physical data qubits under the five-qubit code. Allocating one syndrome and one flag ancilla to every logical block simultaneously produces the conceptual estimate

$$40 + 16 = 56 \quad (89)$$

active qubits. This number is not an exact compiled requirement. Ancillas may be reused sequentially, physical connectivity can introduce routing overhead, and specific logical-gate gadgets may require a different number of temporary qubits. The final hardware qubit count must therefore be obtained after compilation to the target topology.

Hardware-validation strategy. Directly executing the full conceptual FT classifier on hardware is not the most informative first validation step. The proposed hardware study should instead separate three questions. First, the original trained eight-qubit logical inference circuit should be compiled and executed with finite shots to establish the hardware performance of the unencoded VQC. Second, a small representative flagged stabilizer-extraction circuit should be executed independently to verify whether the qualitative syndrome and flag behavior observed in simulation is also measurable on hardware. Third, selected encoded or partially encoded logical operations can be investigated only after the physical-gate counts and topology are known.

Finite-shot inference. All simulator accuracy results in the present study use expectation values obtained from statevector probabilities or Monte Carlo noise trajectories. Hardware execution will instead estimate expectation values from a finite number of measurement shots. For qubit q , the measured feature will take the form

$$\langle \widehat{Z}_q \rangle = \frac{N_0 - N_1}{N_0 + N_1}, \quad (90)$$

where N_0 and N_1 are the observed counts of zero and one outcomes. Shot noise therefore introduces an additional source of uncertainty that must be evaluated before comparing hardware and simulator classification accuracy.

Hardware subset selection. Executing all 10,000 CIFAR-10 test images on a QPU would require a large number of circuit submissions and shots. The hardware study should therefore use a stratified subset containing examples from all ten classes. The subset should be fixed before hardware execution so that simulator and QPU predictions can be compared on exactly the same examples. The full 10,000-sample results should remain the primary simulator evaluation, while hardware runs should be presented as a smaller validation study.

Compilation and connectivity. The eight-logical-qubit VQC contains a ring of nearest-neighbor CNOTs,

$$0 \rightarrow 1 \rightarrow 2 \rightarrow \cdots \rightarrow 7 \rightarrow 0. \quad (91)$$

On a physical processor, the actual cost depends on whether these logical interactions align with the device coupling graph. If additional routing operations are inserted, the effective two-qubit-gate count and circuit depth can increase substantially. Therefore, the hardware experiment should record both the logical circuit statistics and the transpiled physical circuit statistics rather than comparing hardware performance using only the original gate count.

Rigetti/Amazon Braket translation. The next stage of this project is to construct a hardware-preparation pipeline using Amazon Braket and a Rigetti QPU. The pipeline should first reproduce the eight-qubit inference circuit on a shot-based simulator using the same trained parameters and selected CIFAR-10 subset. Once that result is verified, the identical logical circuit should be transpiled to the available Rigetti topology. The hardware stage should report the mapped qubits, circuit depth, one- and two-qubit gate counts, shot count, number of circuit executions, and estimated monetary cost before any large job is submitted.

Separation of simulation and hardware claims. The full encoded FT results in this report should remain simulation results. A small hardware experiment should not be described as an experimental realization of the complete 56-qubit fault-tolerant classifier unless the entire encoded protocol is physically implemented. Instead, hardware results should be clearly labeled according to

what is actually executed, such as *unencoded eight-qubit classifier validation*, *flagged stabilizer-circuit validation*, or *selected encoded-gadget validation*. This separation is necessary to preserve a clear distinction between the logical FT model and the physical experiment.

Future research directions. The results suggest several direct extensions: reducing the QEC-induced error floor through improved syndrome circuits; using Pauli-frame recovery; retaining asymmetric logical Pauli channels; optimizing the number and placement of EC cycles; replacing unconditional error correction with adaptive scheduling; redesigning the variational ansatz to reduce non-Clifford operations; and investigating codes or logical-gate sets better matched to the classifier. A particularly important direction is joint optimization of the quantum machine-learning circuit and the fault-tolerant architecture, rather than treating them as two independent design problems.

13 Conclusion

Summary. This work developed a progressive simulation framework for studying fault-tolerant quantum machine learning using a ten-class CIFAR-10 hybrid quantum classifier. The study began with basic three-qubit bit-flip correction and progressively introduced noisy syndrome extraction, repeated syndrome measurements, fault propagation, transversal operations, general depolarizing noise, the $[[5, 1, 3]]$ perfect code, flagged stabilizer extraction, temporal syndrome decoding, and logical-gate overhead. The progression allowed the contribution of each fault-tolerance assumption to be examined separately rather than treating QEC as an ideal black box.

Ideal classifier and physical-noise sensitivity. The trained eight-logical-qubit classifier achieved

$$91.43\% \tag{92}$$

test accuracy and a Macro-F1 score of

$$91.46\% \tag{93}$$

under ideal quantum simulation. The unprotected circuit degraded rapidly as physical Pauli faults accumulated across its many operation locations. This established a sufficiently sensitive application-level benchmark for evaluating whether error correction preserved useful computation rather than merely reducing an abstract logical error metric.

Ideal QEC result. Under ideal $[[5, 1, 3]]$ protection, general Pauli errors were strongly suppressed. At $p = 0.01$, the exact logical error probability was reduced to

$$p_L = 0.000978, \tag{94}$$

and the corresponding classifier retained 91.30% accuracy compared with 74.74% for the unprotected model. At $p = 0.02$, ideal QEC maintained 90.64% accuracy while the unprotected classifier fell to 34.43%. These experiments established that sufficiently low logical error rates could preserve the original machine-learning model.

Noisy QEC result. When syndrome extraction became noisy, the distinction between QEC and FTQEC became explicit. Naive single-ancilla extraction produced a logical-error floor of

$$3.9094\% \tag{95}$$

even at zero externally applied data noise and reduced CIFAR-10 accuracy to approximately 15.10%. Introducing a flag ancilla reduced this floor to

$$1.3274\%, \quad (96)$$

corresponding to an approximately 66% reduction in QEC-induced logical error and increasing the corresponding classification accuracy to 59.21%. The result demonstrates that controlling fault propagation can be as important as reducing the underlying physical error rate.

Temporal-decoding result. Three-round syndrome-history decoding outperformed bitwise majority voting because temporal syndrome changes contained information about faults occurring between rounds. Across the investigated physical-noise sweep, the mean logical error decreased from

$$0.083861 \quad (97)$$

for raw majority voting to

$$0.062830 \quad (98)$$

for history decoding. However, both remained worse than the adaptive flagged strategy, whose mean logical error was 0.023167. Thus, acquiring additional syndrome information was beneficial only when the information gain justified the additional noisy circuitry required to obtain it.

Final gate-aware result. The adaptive flagged decoder was therefore selected for the final end-to-end model. The classifier contains 96 arbitrary rotations and 32 logical CNOTs, corresponding to 160 logical-qubit fault locations in the uniform model and 320 EC-equivalent protected cycles under the two-cycle gate-aware overhead model. Even at $p = 0$, the final model predicted approximately

$$4.21 \quad (99)$$

non-identity logical Pauli events per inference. Consequently, the UniformProtected model achieved 58.90% accuracy and the final gate-aware FT model achieved only

$$23.52\%. \quad (100)$$

No FinalFT/unprotected break-even point was observed for

$$0 \leq p \leq 0.05, \quad (101)$$

and the FinalFT model did not reach the target 77% classification accuracy at any tested physical error probability.

Main conclusion. The results show that ideal error correction alone substantially overestimates the robustness of a protected quantum machine-learning model. For a useful application-level advantage, the logical error rate after noisy fault-tolerant correction must be sufficiently small relative to the total number of protected operations. In the present configuration, the $[[5, 1, 3]]$ code successfully suppresses isolated physical Pauli errors, and flagged extraction substantially reduces dangerous fault propagation, but the remaining correction and logical-gate overhead accumulates too rapidly across the classifier.

QEC becomes application-level fault tolerance only when $N_{\text{logical}} p_L \ll 1$.

(102)

Implication. The negative final break-even result should therefore not be interpreted as evidence that fault-tolerant quantum machine learning is inherently ineffective. Instead, it identifies the engineering requirement that a useful FT architecture must simultaneously reduce the logical error probability, limit the number of expensive correction cycles, and use logical operations compatible with the selected code. The results further suggest that future quantum machine-learning circuits should be designed with fault tolerance as an explicit architectural constraint rather than retrofitting a NISQ-oriented variational circuit after training.